Privacy-Preserving Machine Learning on Web Browsing for Public Opinion

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Traditional Political Polling

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West Virginia 2024 Presidential Election Polls

Harris vs. Trump

Source	Date	Sample	Harris	Trump	Other
Research America	8/30/2024	400 LV ±4.9%	34%	61%	5%

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Michigan 2024 Presidential Election Polls

Instantly compare a poll to prior one by same pollster

Harris vs. Trump

Source	Date	Sample	Harris	Trump	Other
Average of 23 Polls†			**48.6%**	**46.8%**	-
FAU / Mainstreet	11/04/2024	713 LV	49%	47%	4%
Emerson College	11/04/2024	790 LV ±3.4%	50%	48%	2%
Research Co.	11/04/2024	450 LV ±4.6%	49%	47%	4%
InsiderAdvantage	11/03/2024	800 LV ±3.7%	47%	47%	6%
Trafalgar Group	11/03/2024	1,079 LV ±2.9%	47%	48%	5%
MIRS / Mich. News Source	11/03/2024	585 LV ±4%	50%	48%	2%
NY Times / Siena College	11/03/2024	998 LV ±3.7%	47%	47%	6%
Morning Consult	11/03/2024	1,108 LV ±3%	49%	48%	3%
AtlasIntel	11/02/2024	1,198 LV ±3%	48%	50%	2%
Redfield & Wilton	11/01/2024	1,731 LV ±2.2%	47%	47%	6%
The Times (UK) / YouGov	11/01/2024	942 LV ±3.9%	48%	45%	7%
EPIC-MRA	11/01/2024	600 LV ±4%	48%	45%	7%
Marist Poll	11/01/2024	1,214 LV ±3.5%	51%	48%	1%
AtlasIntel	10/31/2024	1,136 LV ±3%	49%	49%	2%
Echelon Insights	10/31/2024	600 LV ±4.4%	48%	48%	4%
MIRS / Mich. News Source	10/31/2024	1,117 LV ±2.5%	47%	49%	4%
UMass Lowell	10/31/2024	600 LV ±4.5%	49%	45%	6%
Washington Post	10/31/2024	1,003 LV ±3.7%	47%	46%	7%
Fox News	10/30/2024	988 LV ±3%	49%	49%	2%
CNN	10/30/2024	726 LV ±4.7%	48%	43%	9%
Suffolk University	10/30/2024	500 LV ±4.4%	47%	47%	6%

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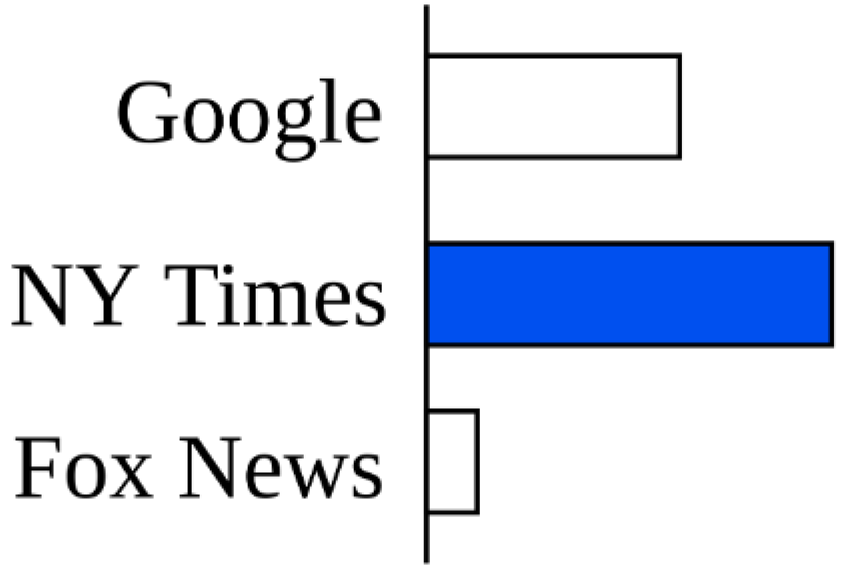

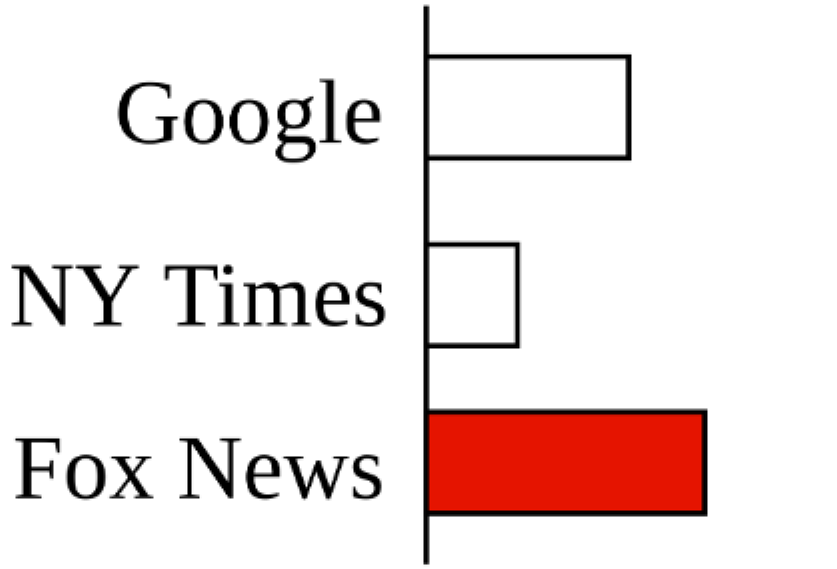

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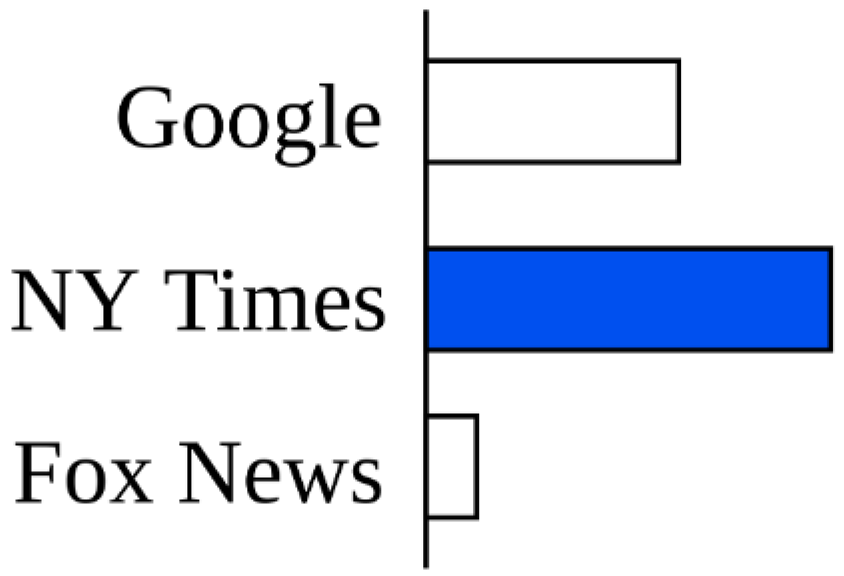

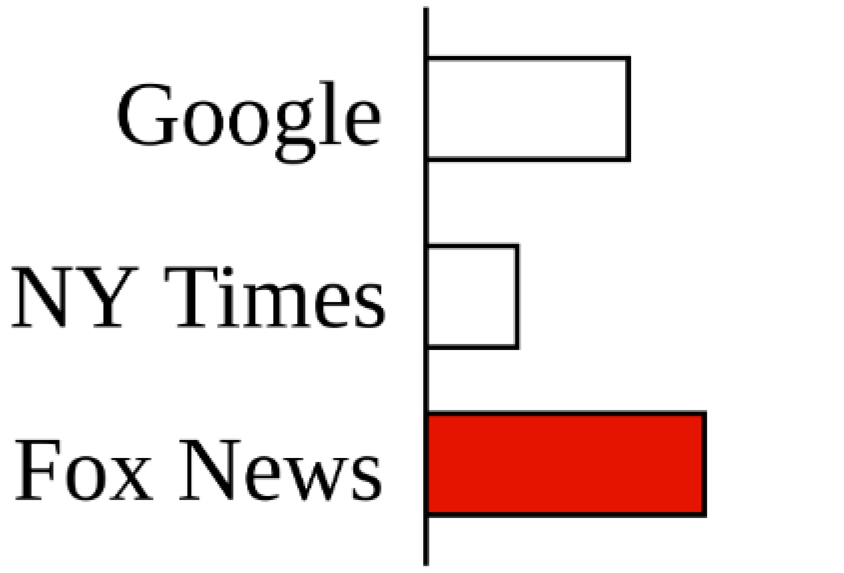

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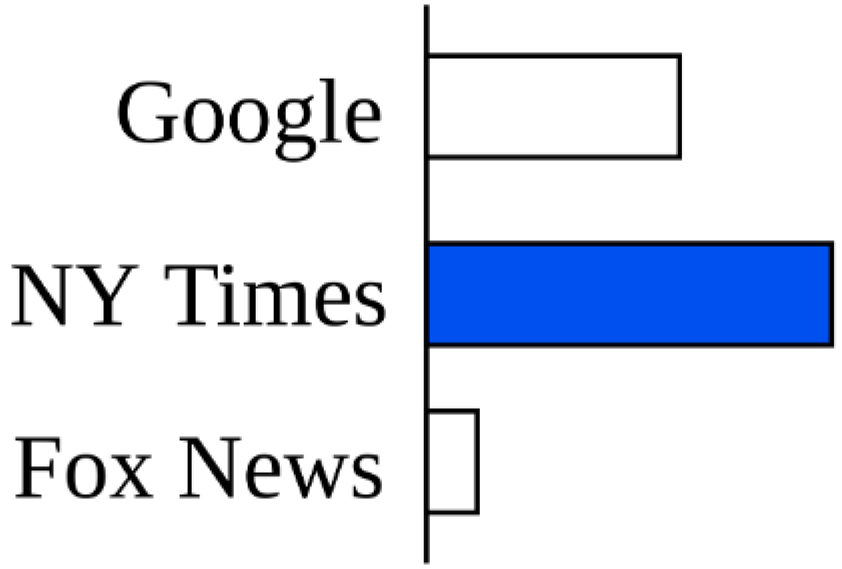
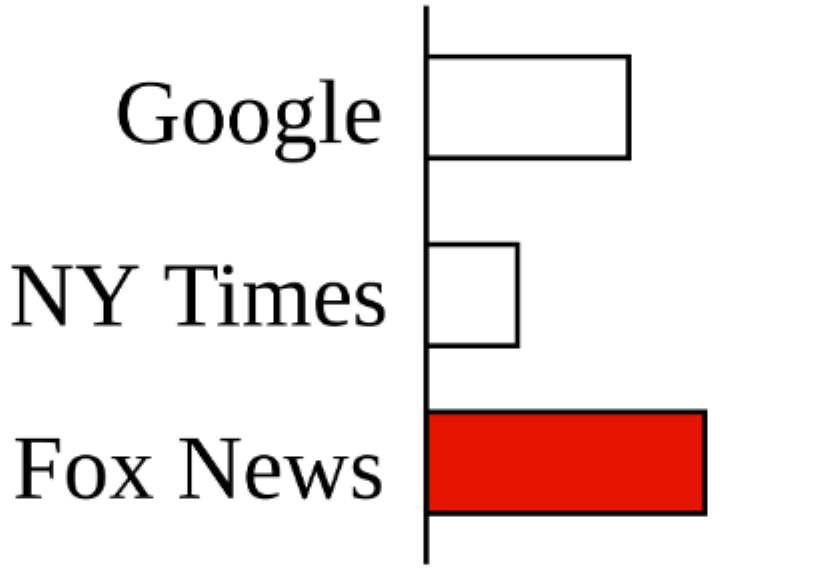
- Example - news websites
- More data
- Fully automated





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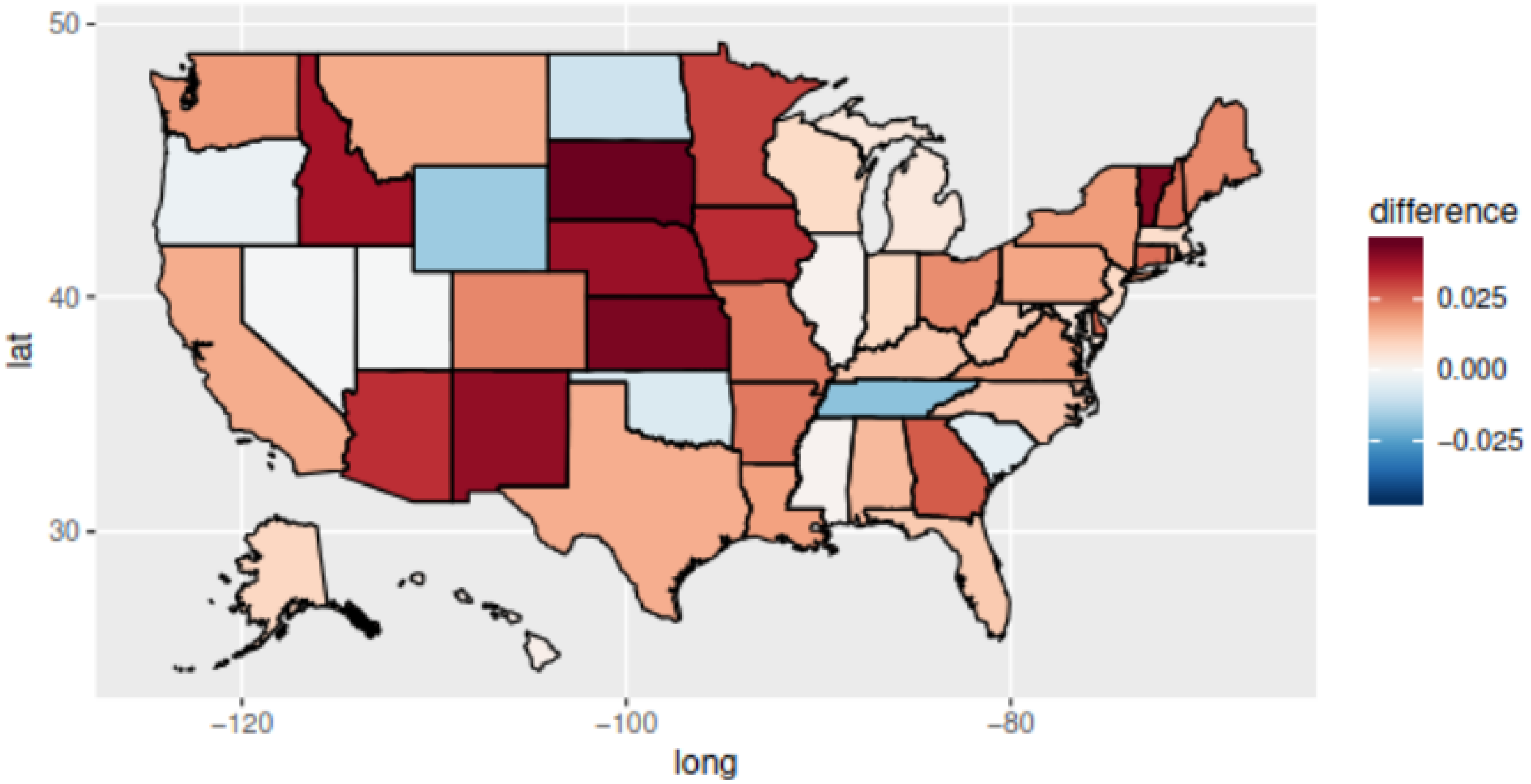

Figure 8: Impact of the 'Comey letter' at the state level.

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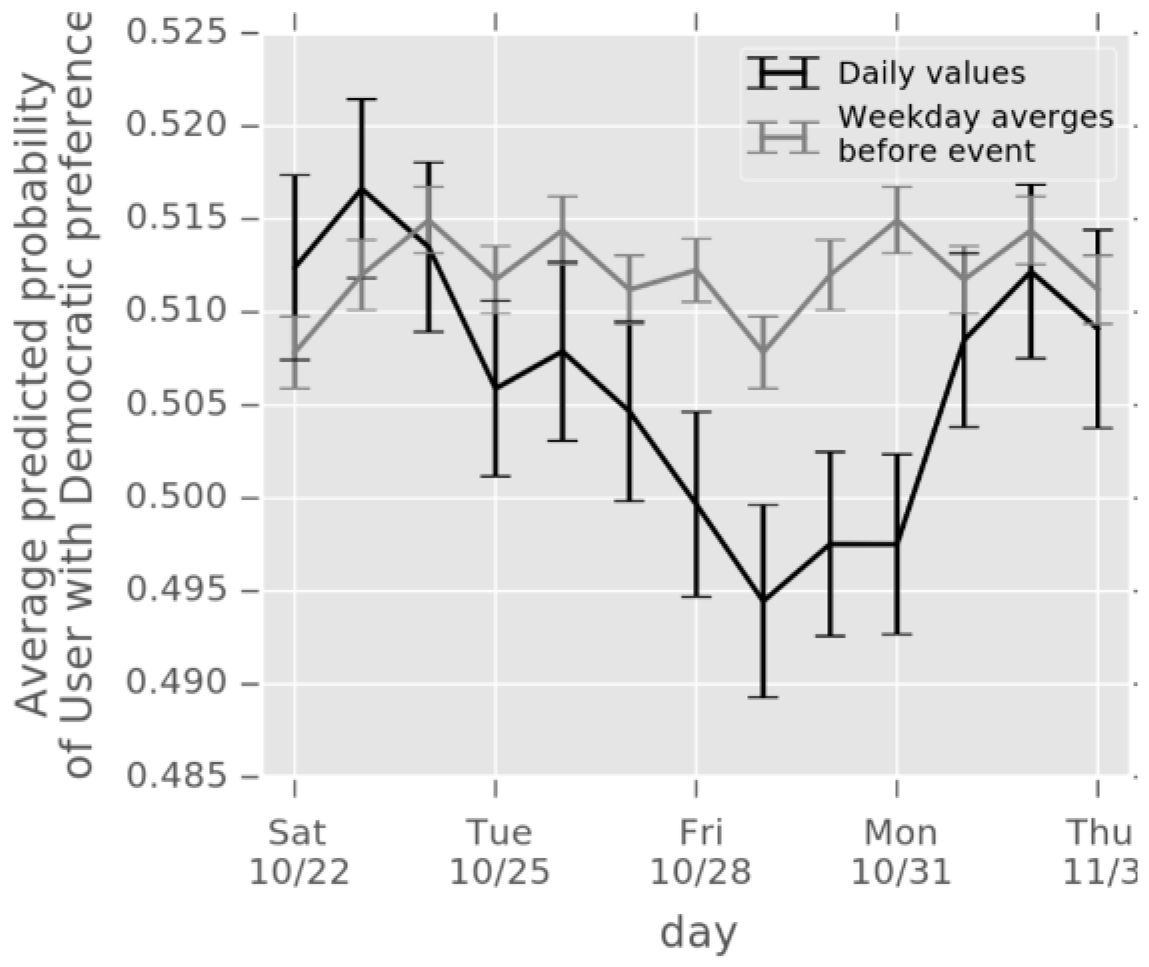

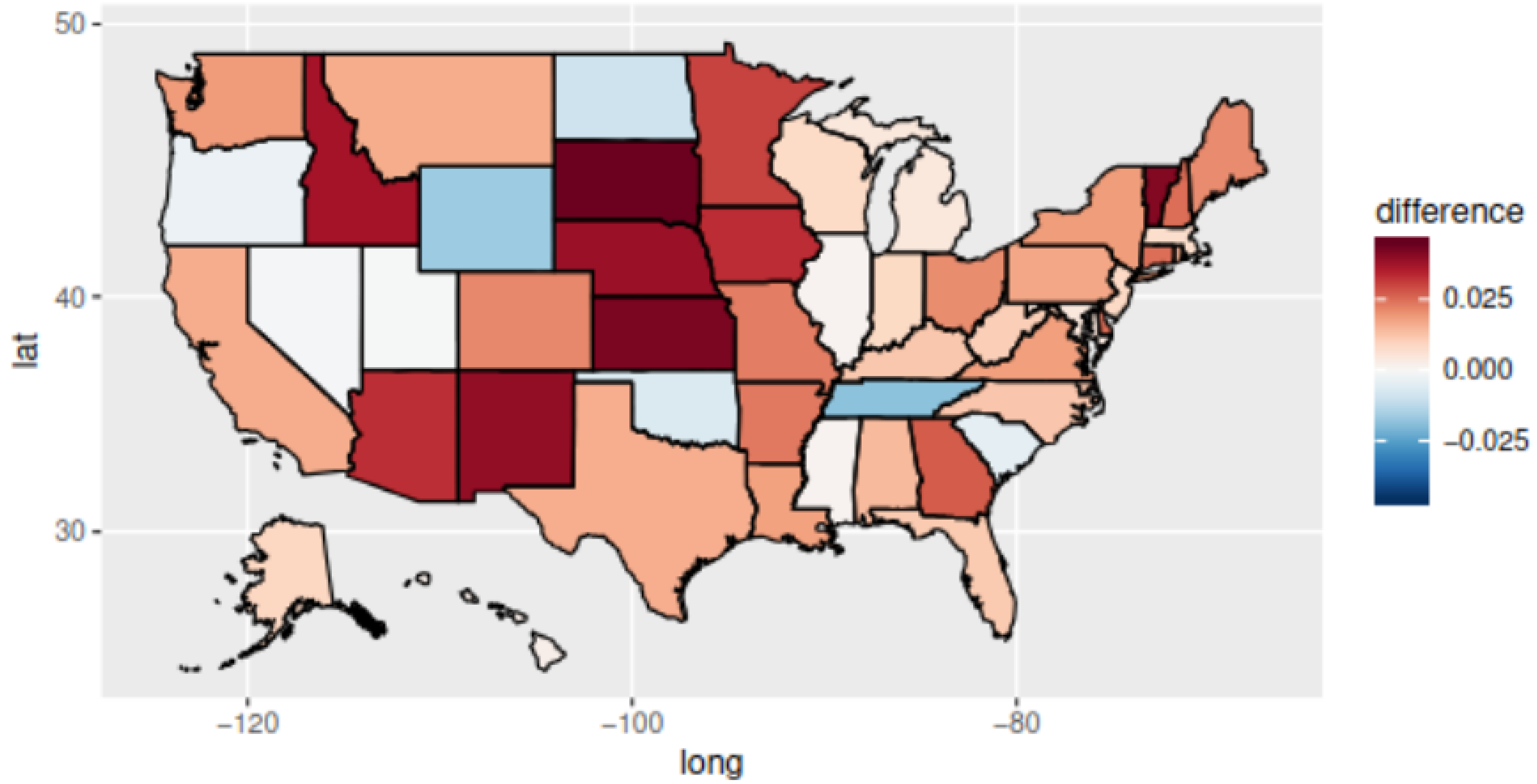

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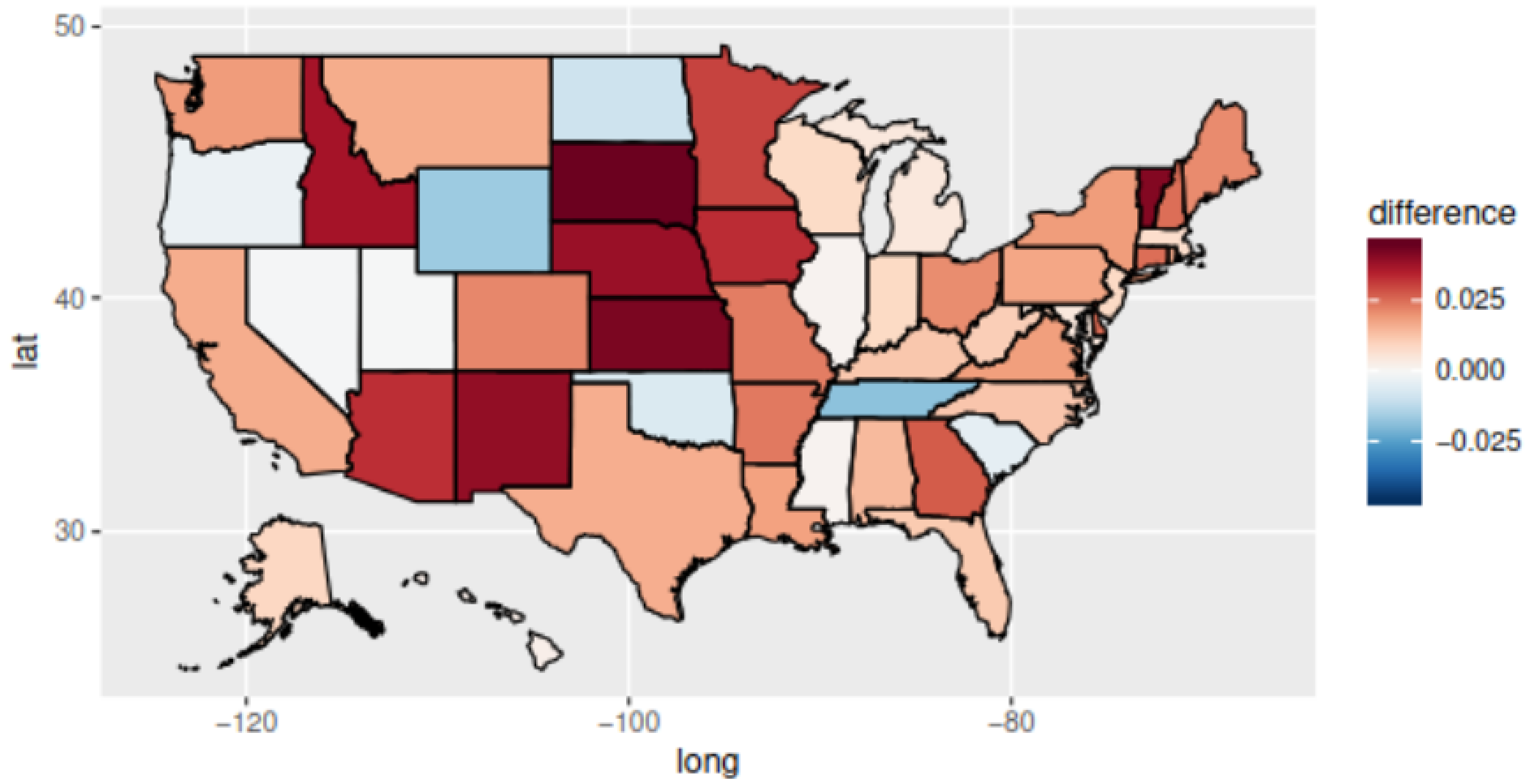

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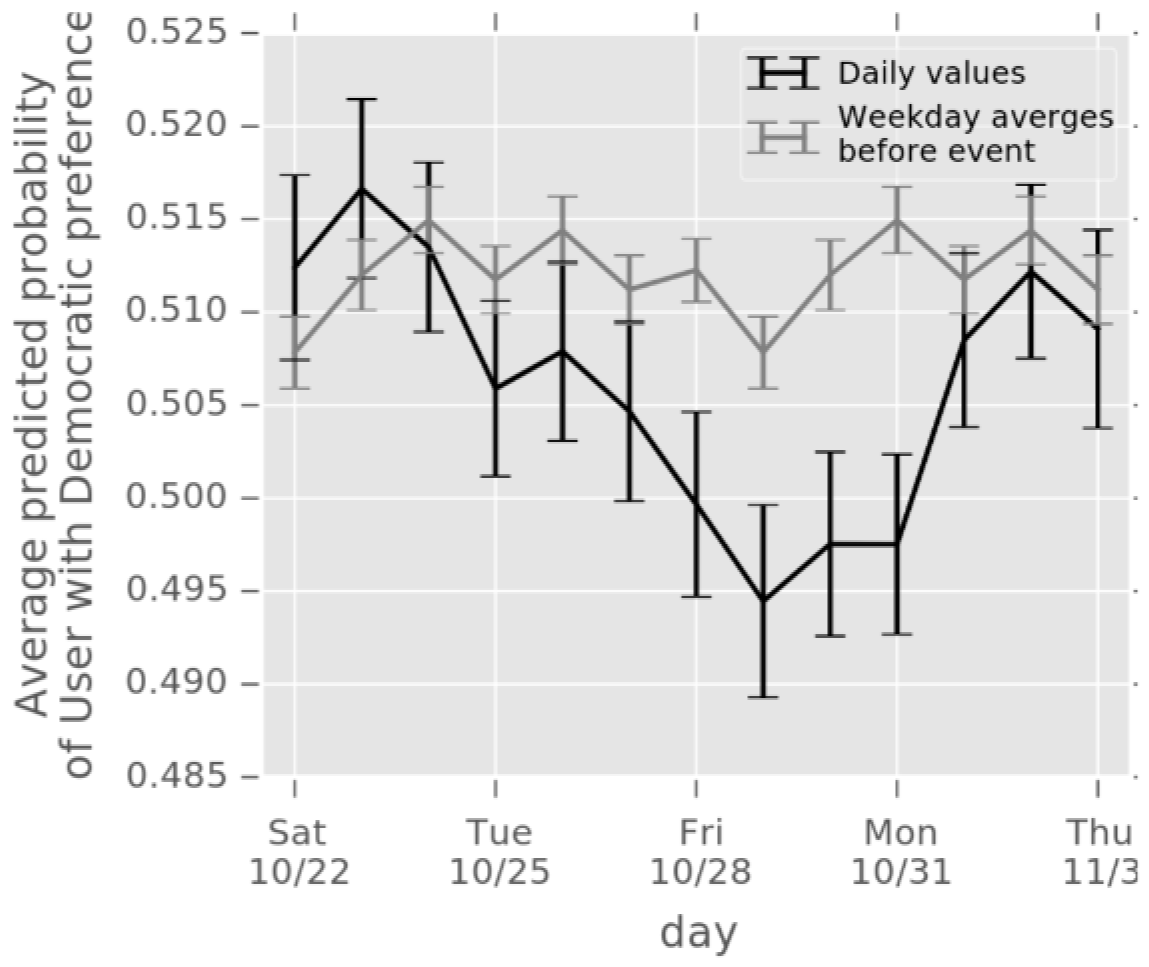

Prior Work

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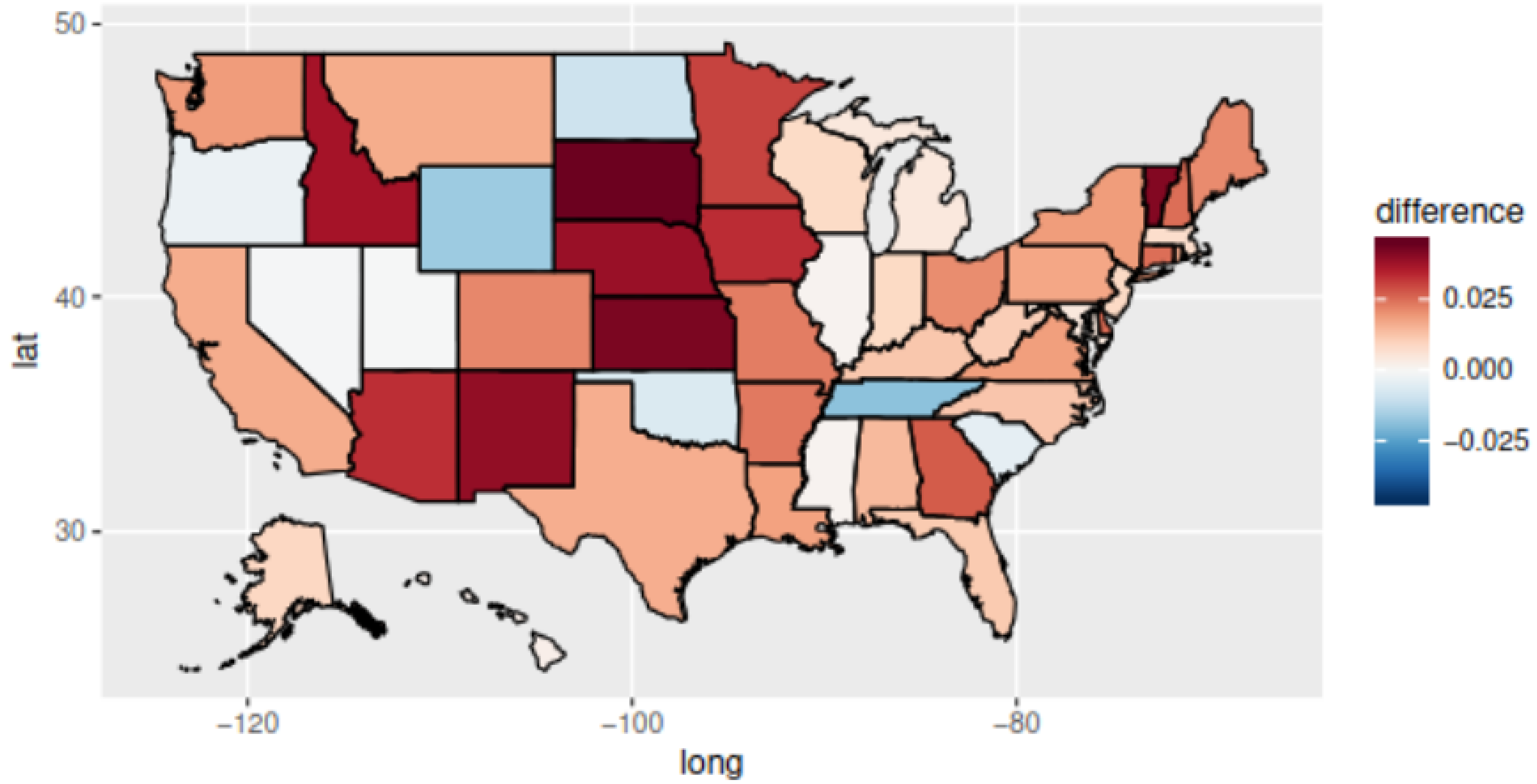
- Quantifying the 'Comey letter' (Comarela et al.)
- Social media referrals are the best signal
- Used large plaintext dataset



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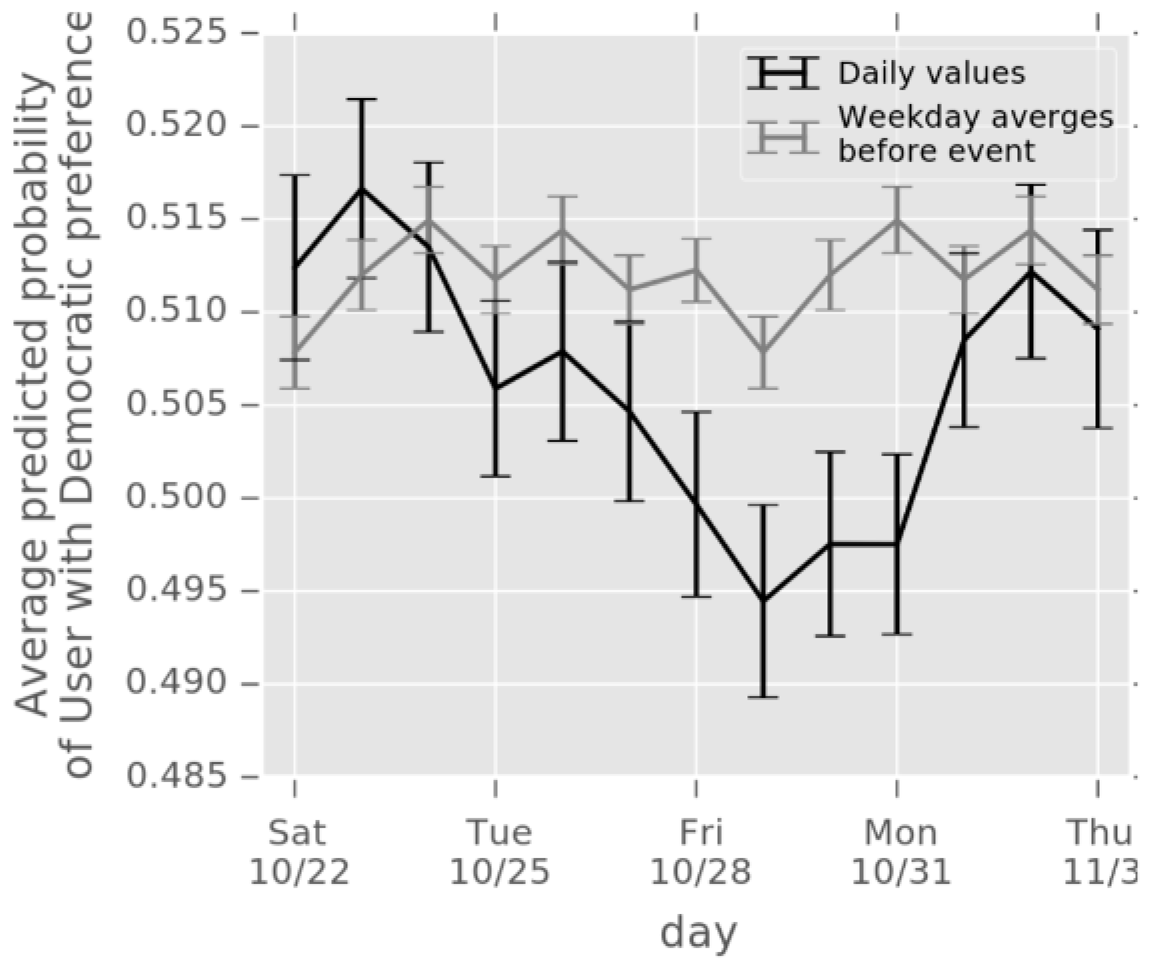

Two Approaches to Political Polling

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- Slow
- Expensive
- Coarse-grained insights

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VS

Web Behavior Analysis

- Immediate
- Cheap
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Two Approaches to Political Polling

Traditional Polling		Web Behavior Analysis
Slow	VS	Immediate
Expensive		Cheap
Coarse-grained insights		Fine-grained insights

What about privacy?

Our Contributions

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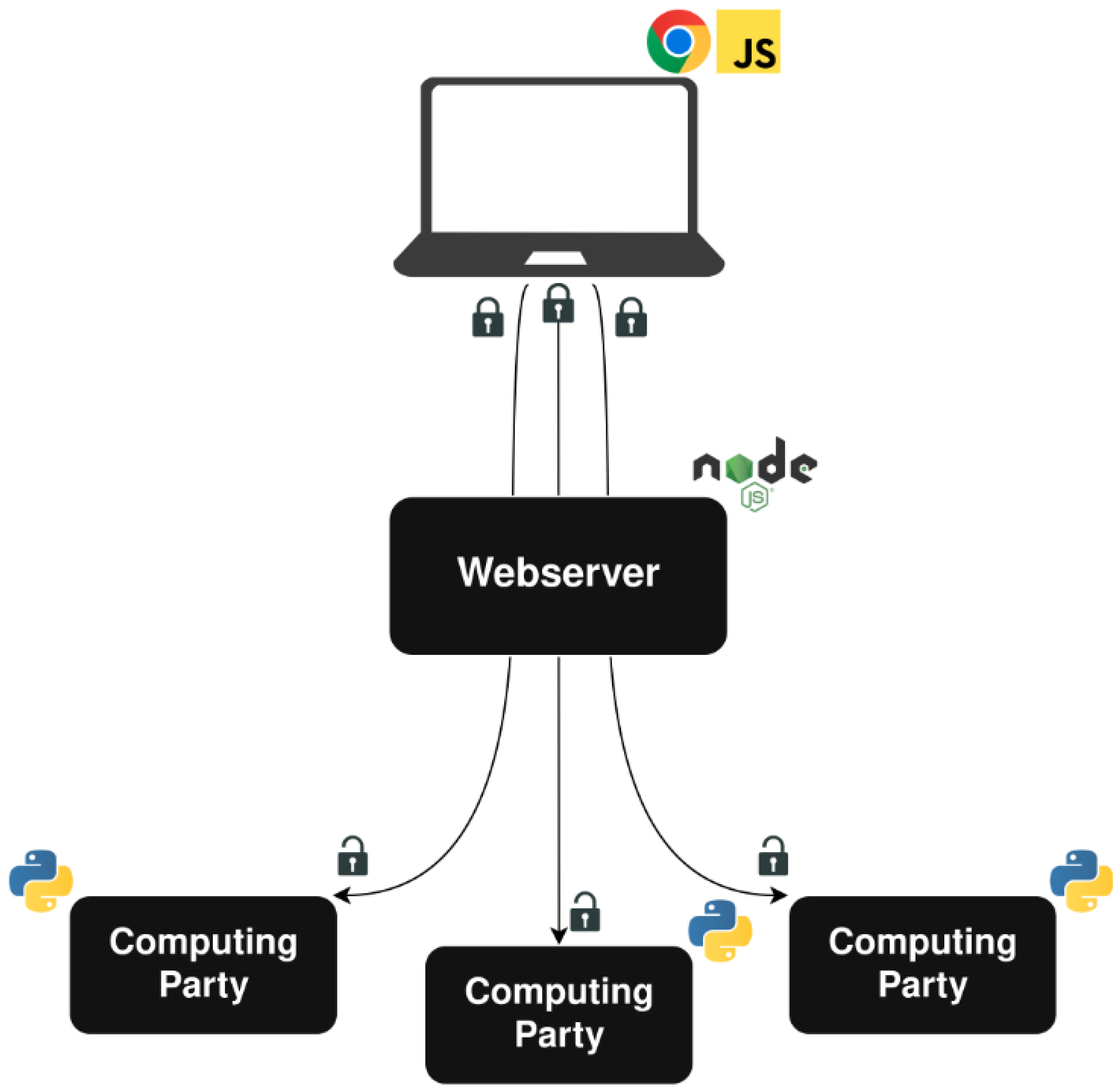
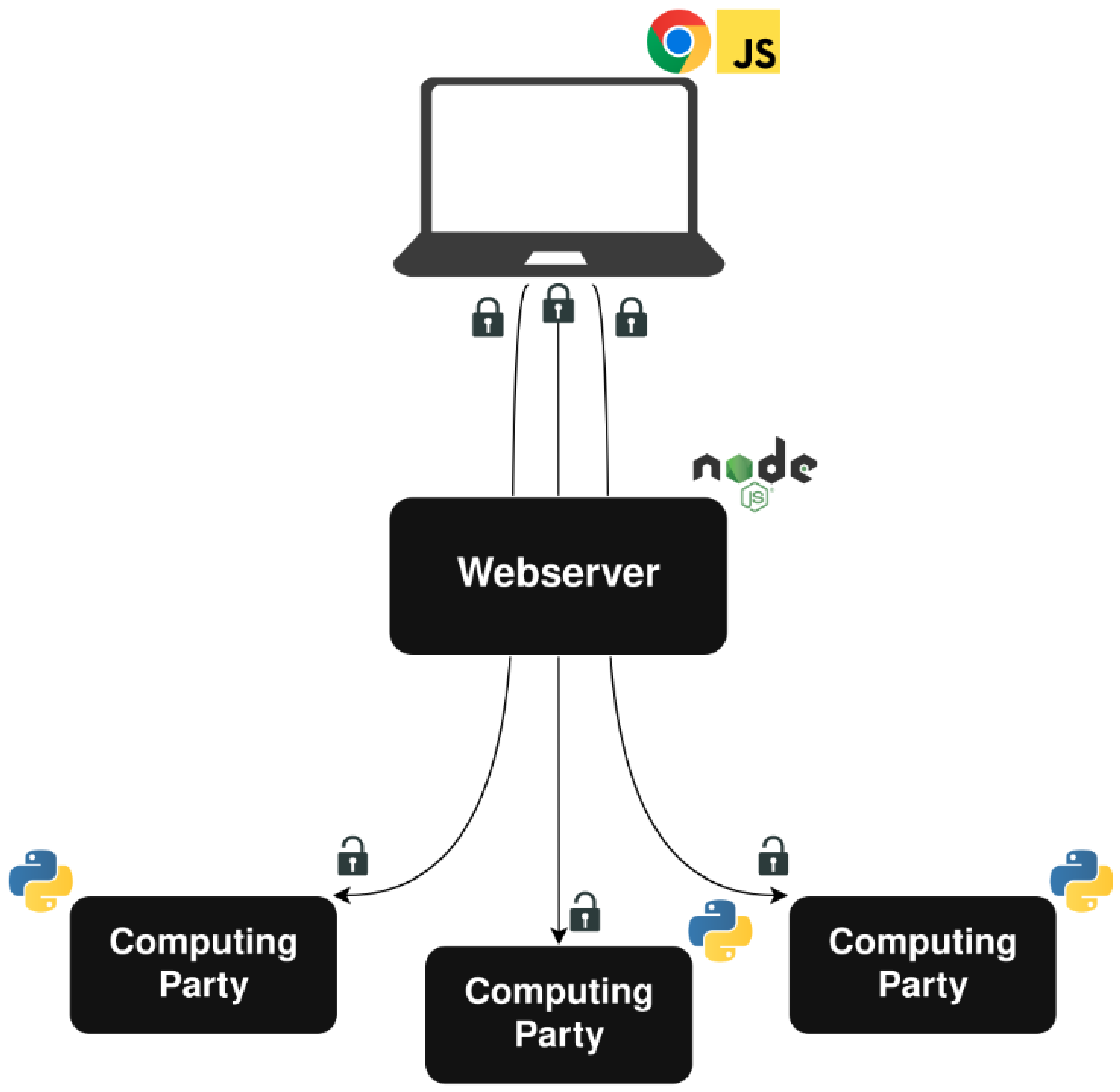
- We collected and analyzed data from almost 8000 unique users
- All analysis took place under MPC

System Design



System Design

Client Plugin



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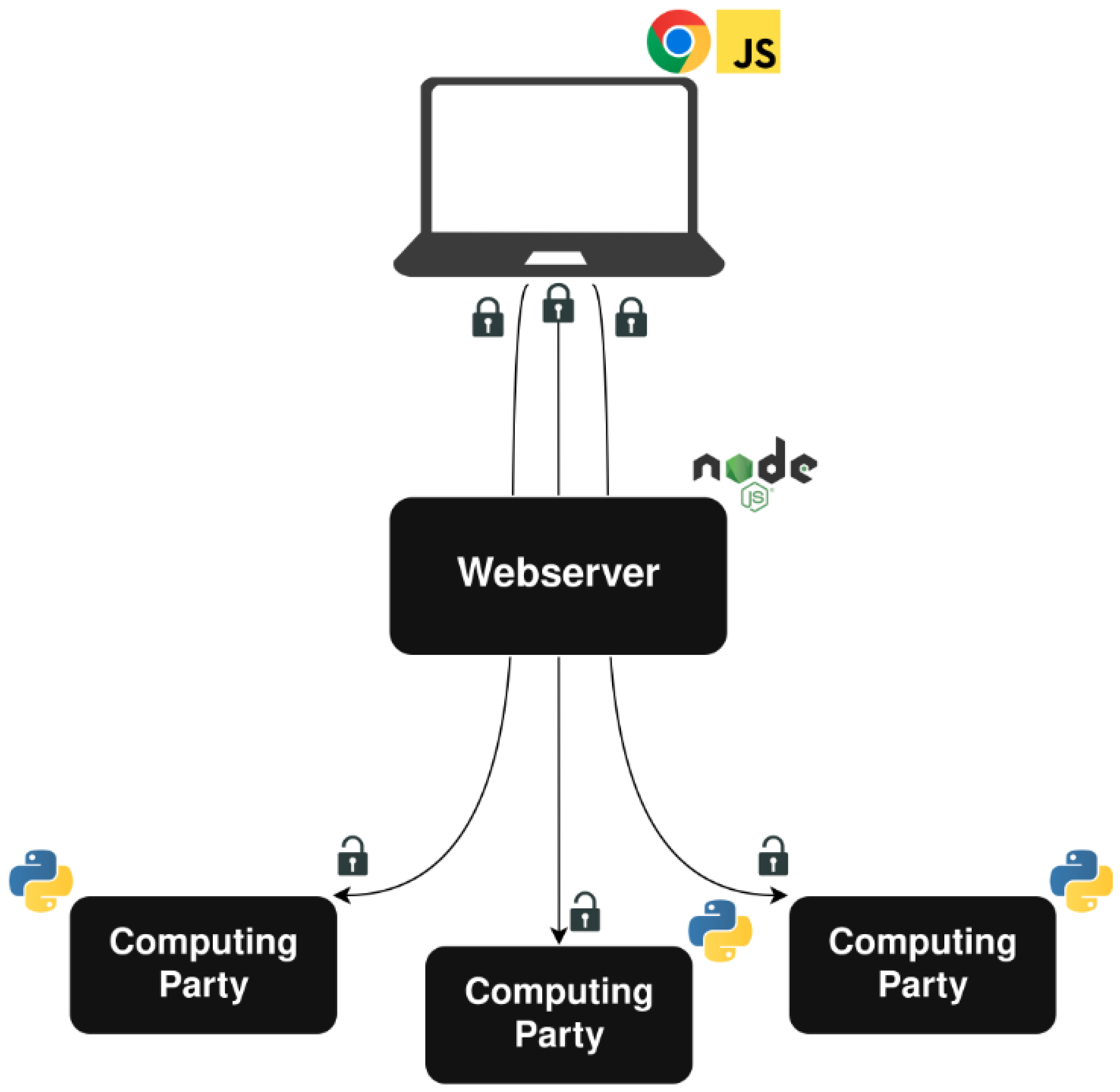

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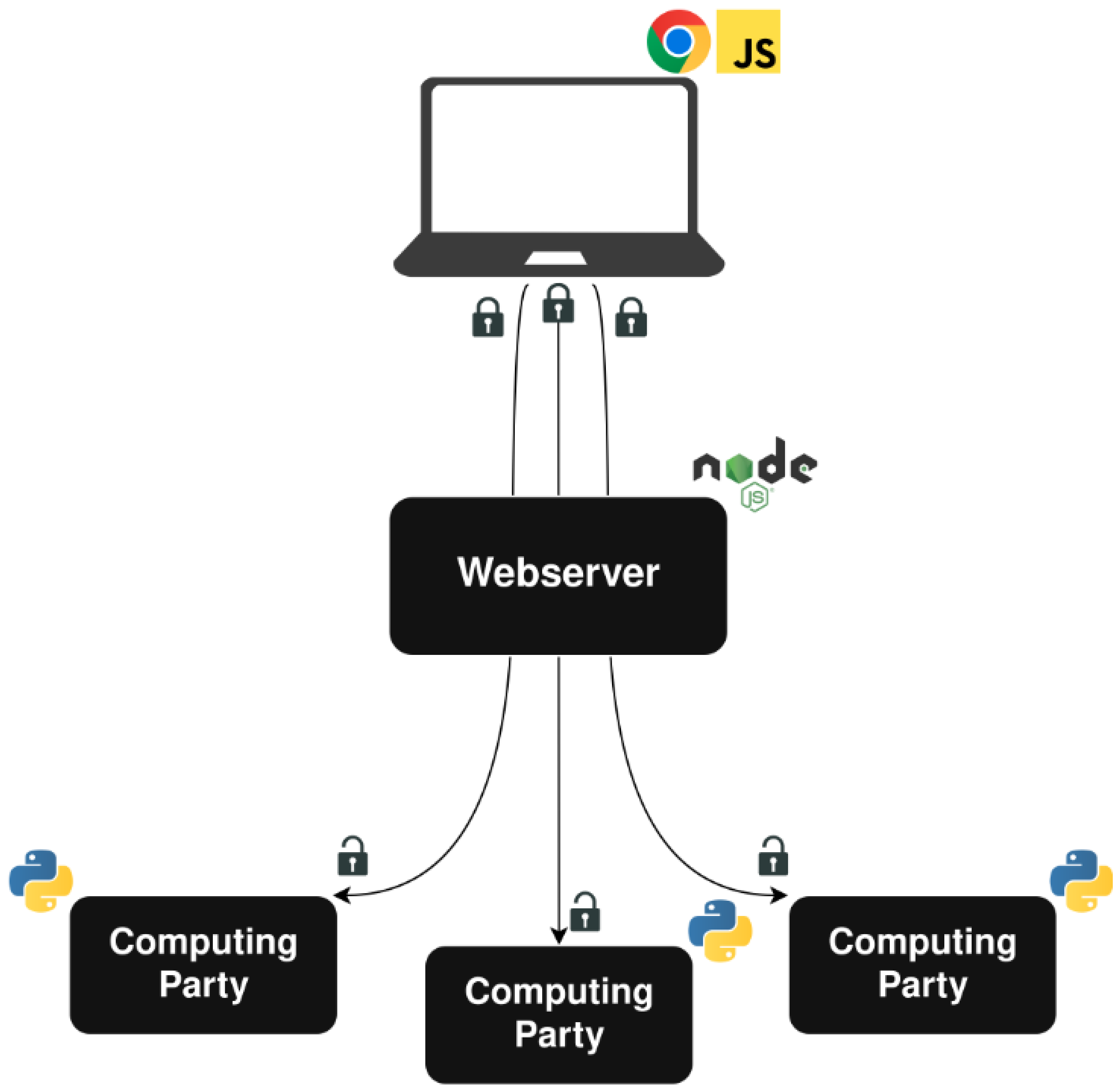

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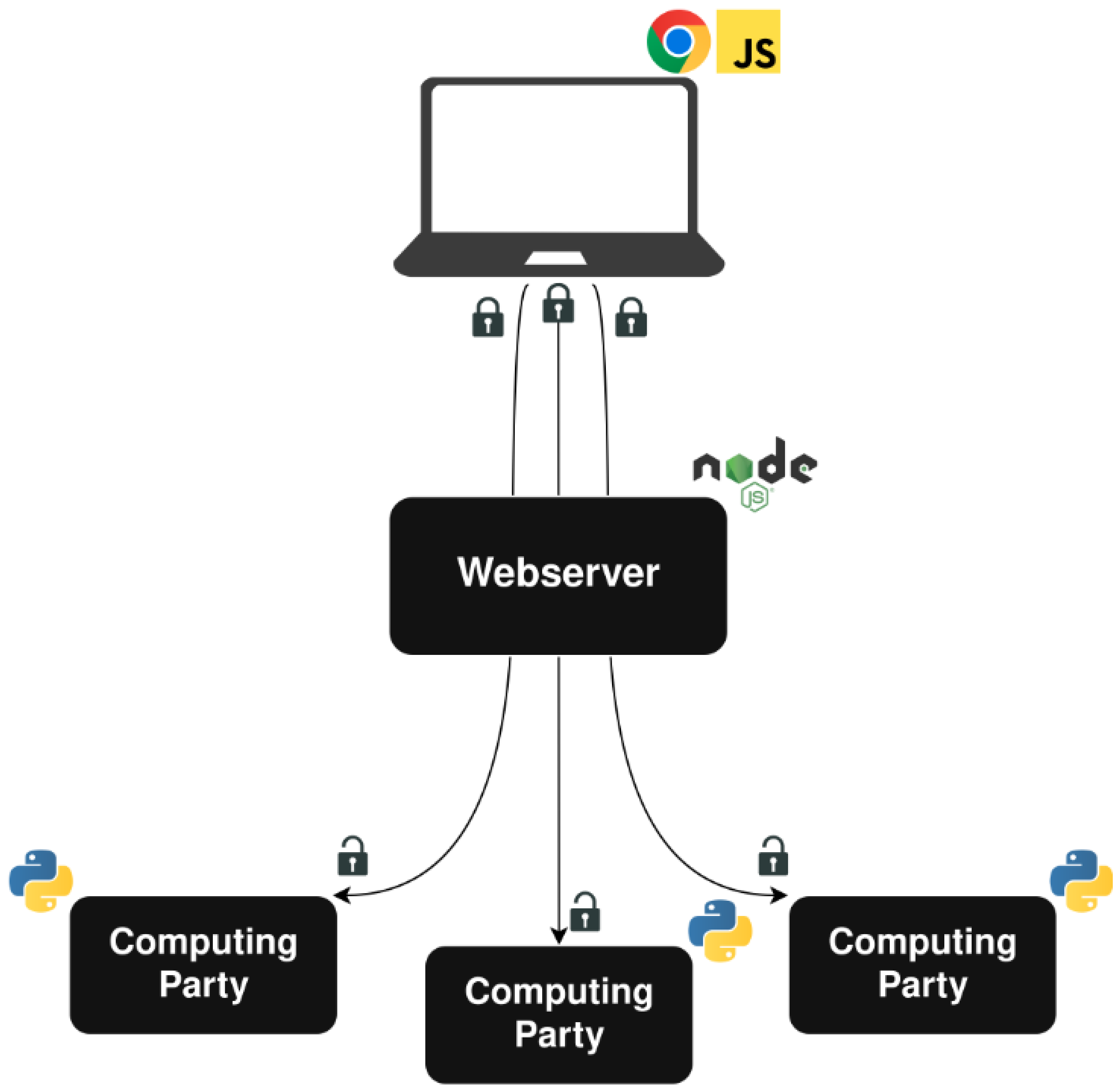

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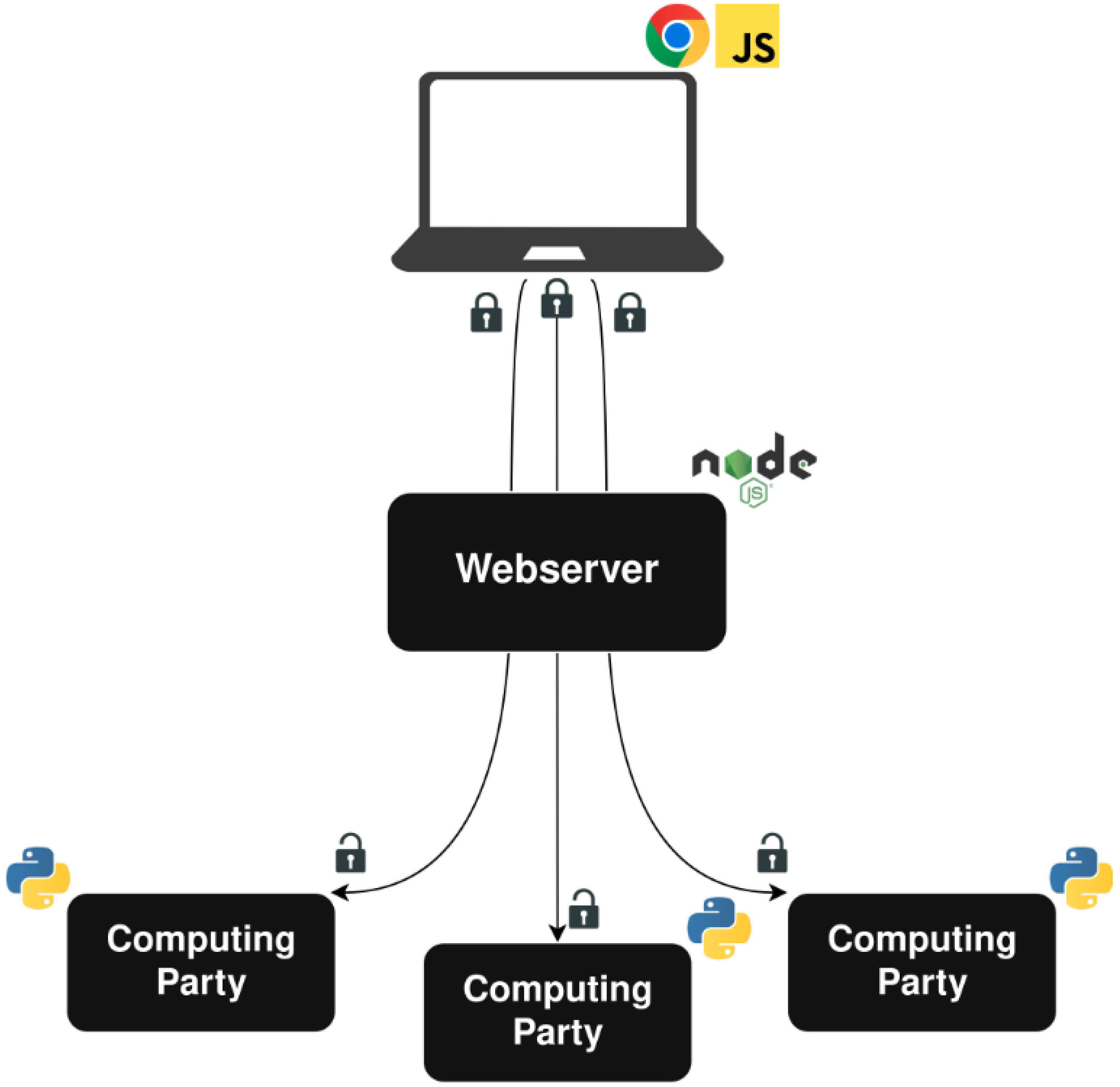

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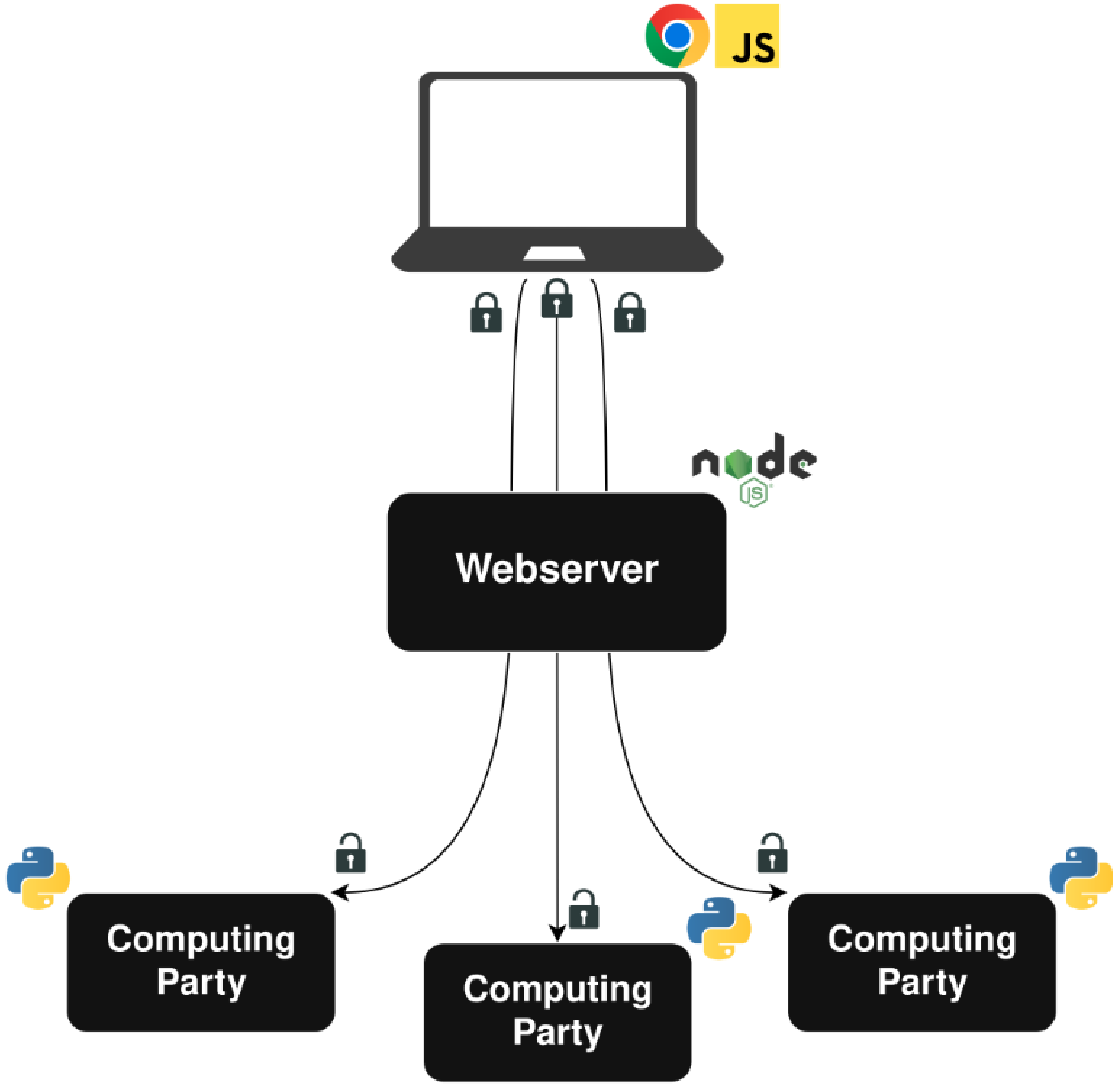

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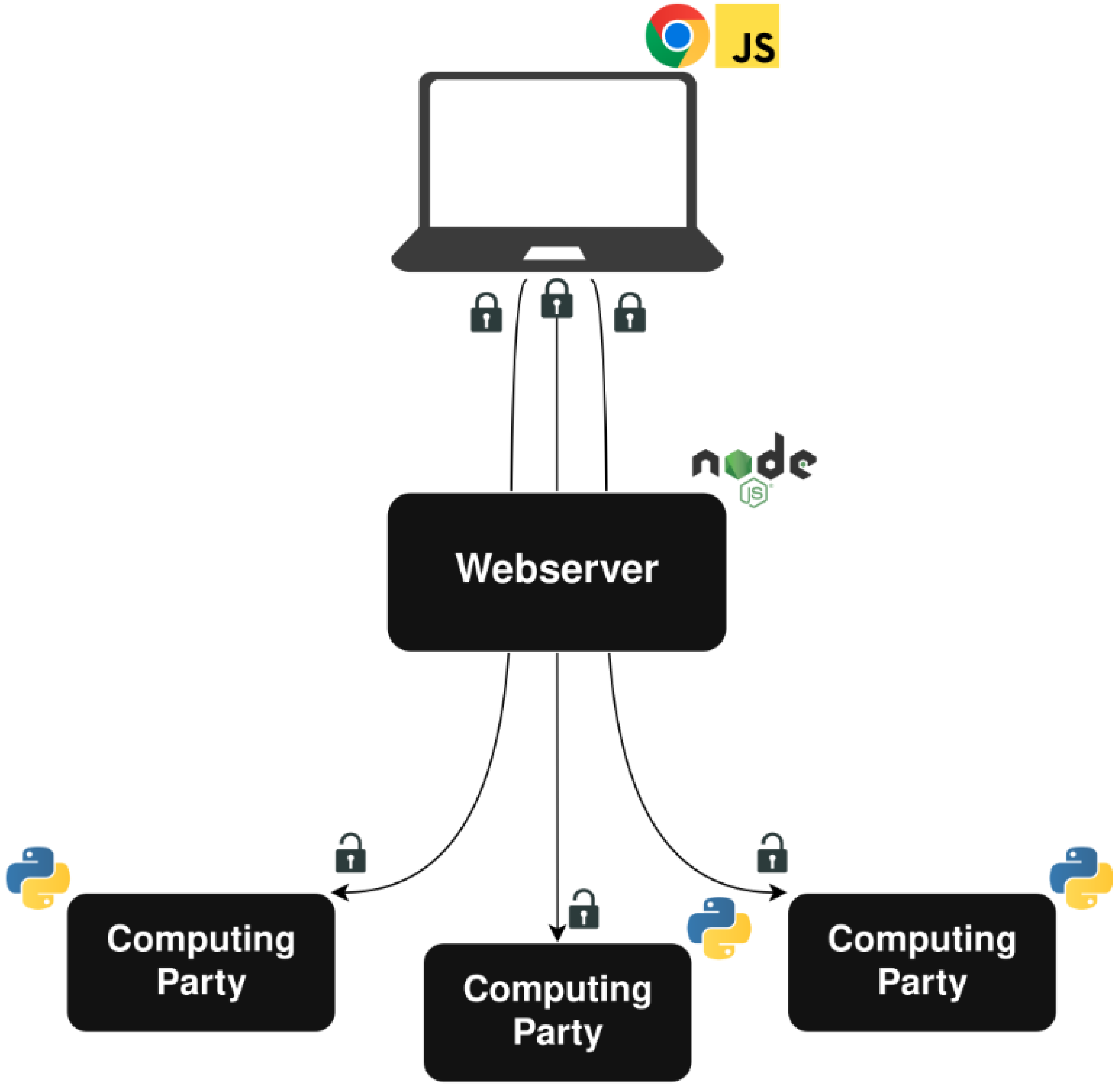

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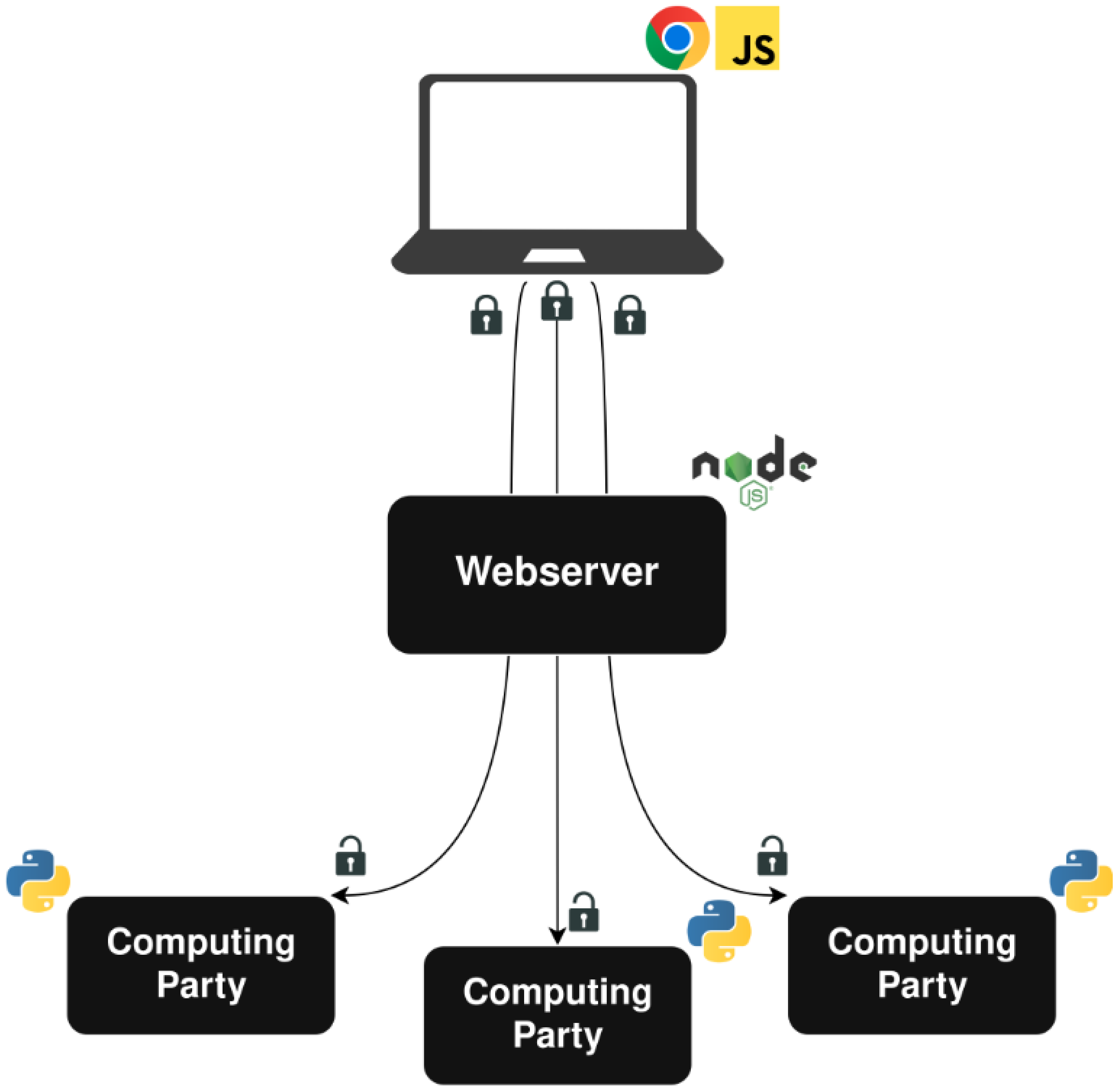

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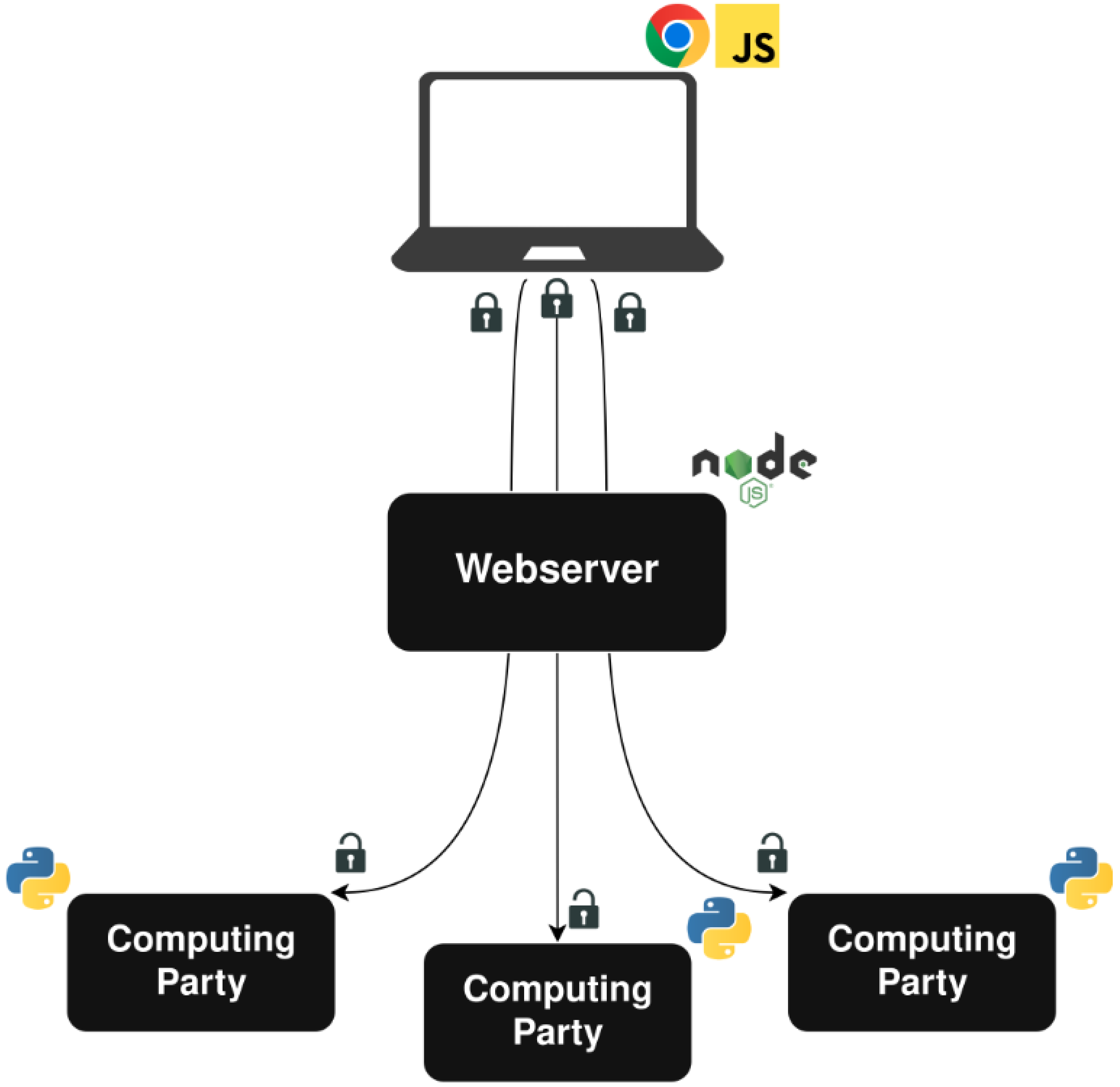
- Implementation is open source



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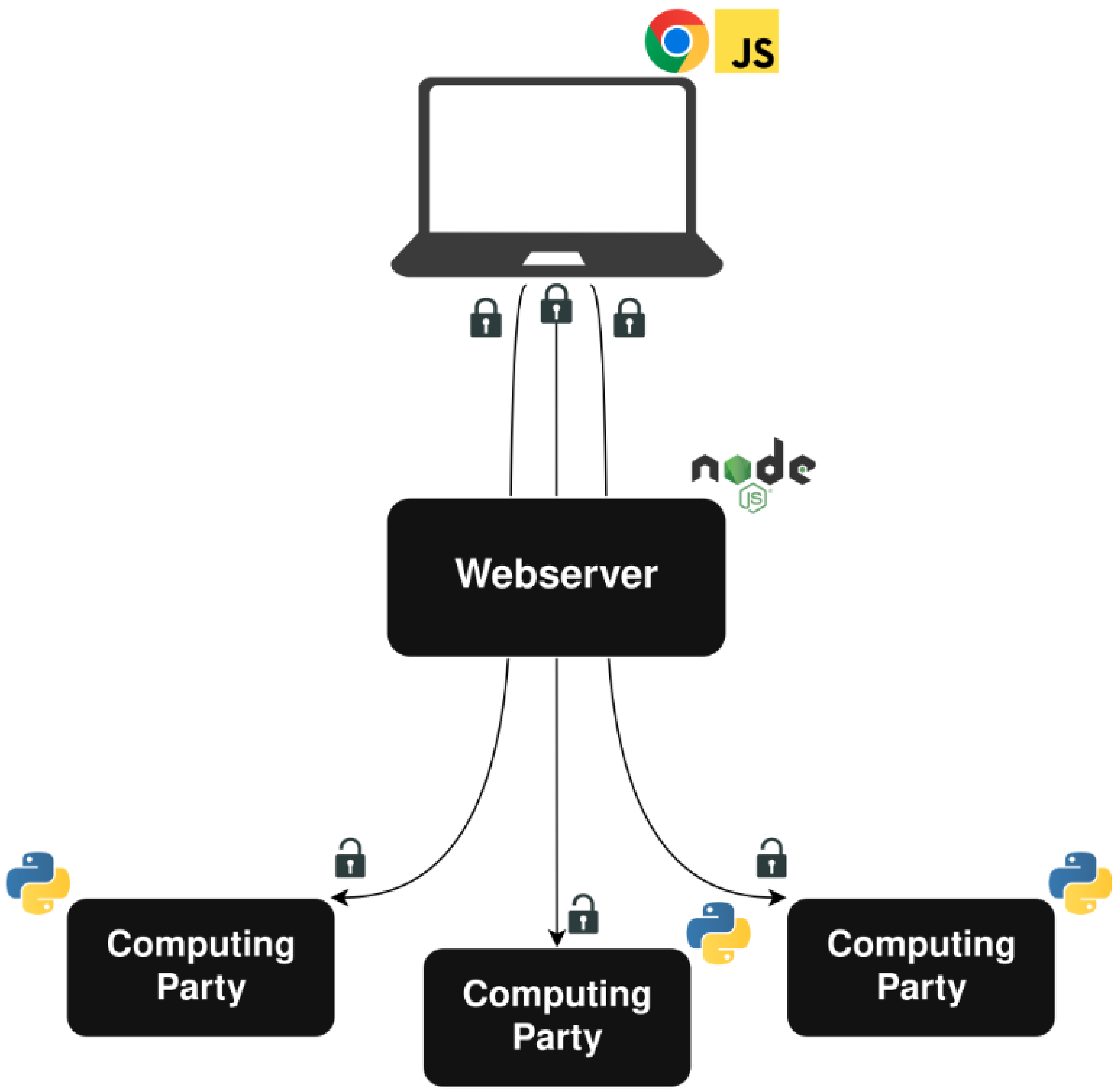
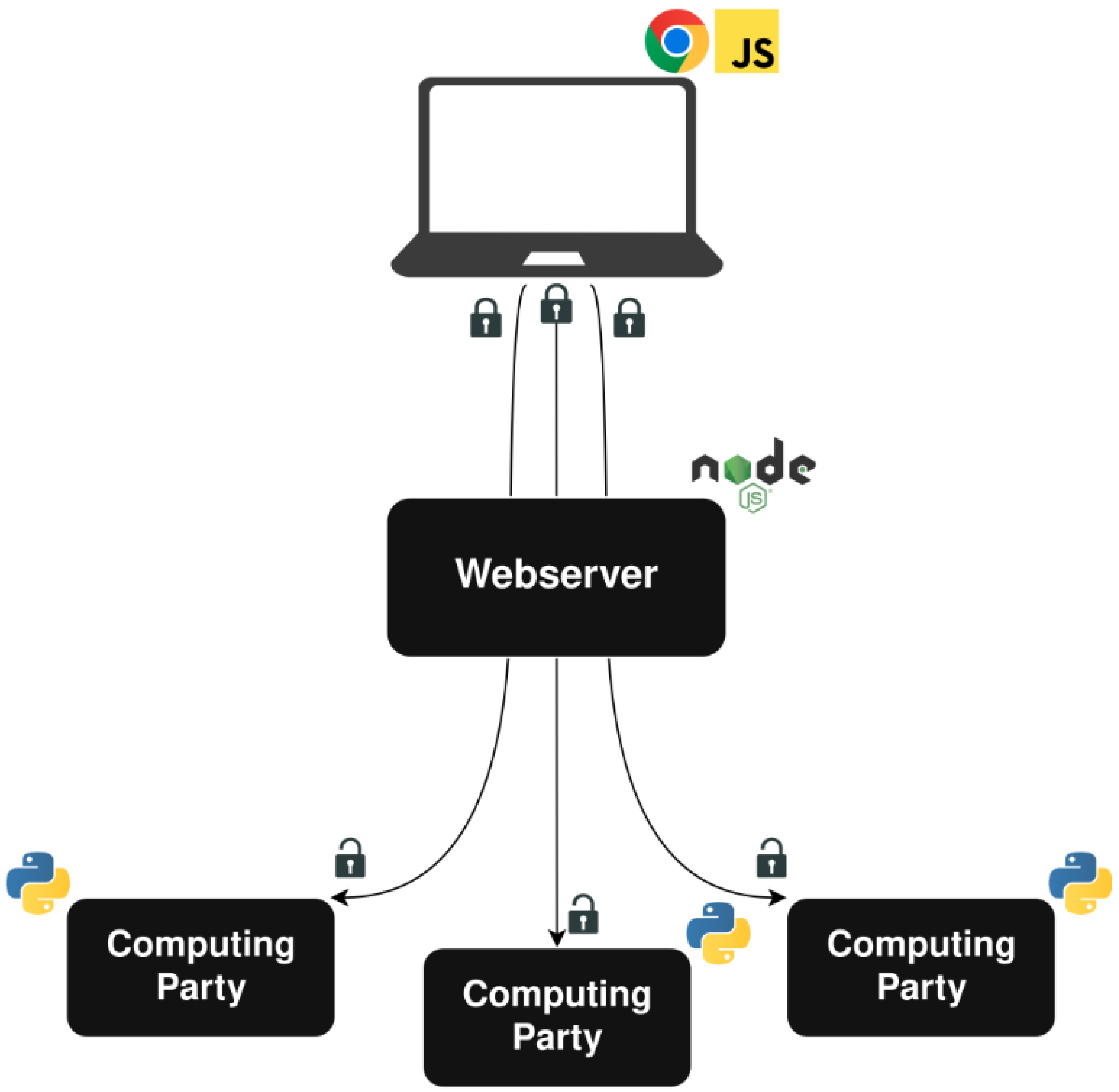


JS
Webserver
Computing Party
Computing Party
Computing Party

Webserver

- Simplifies interaction with clients



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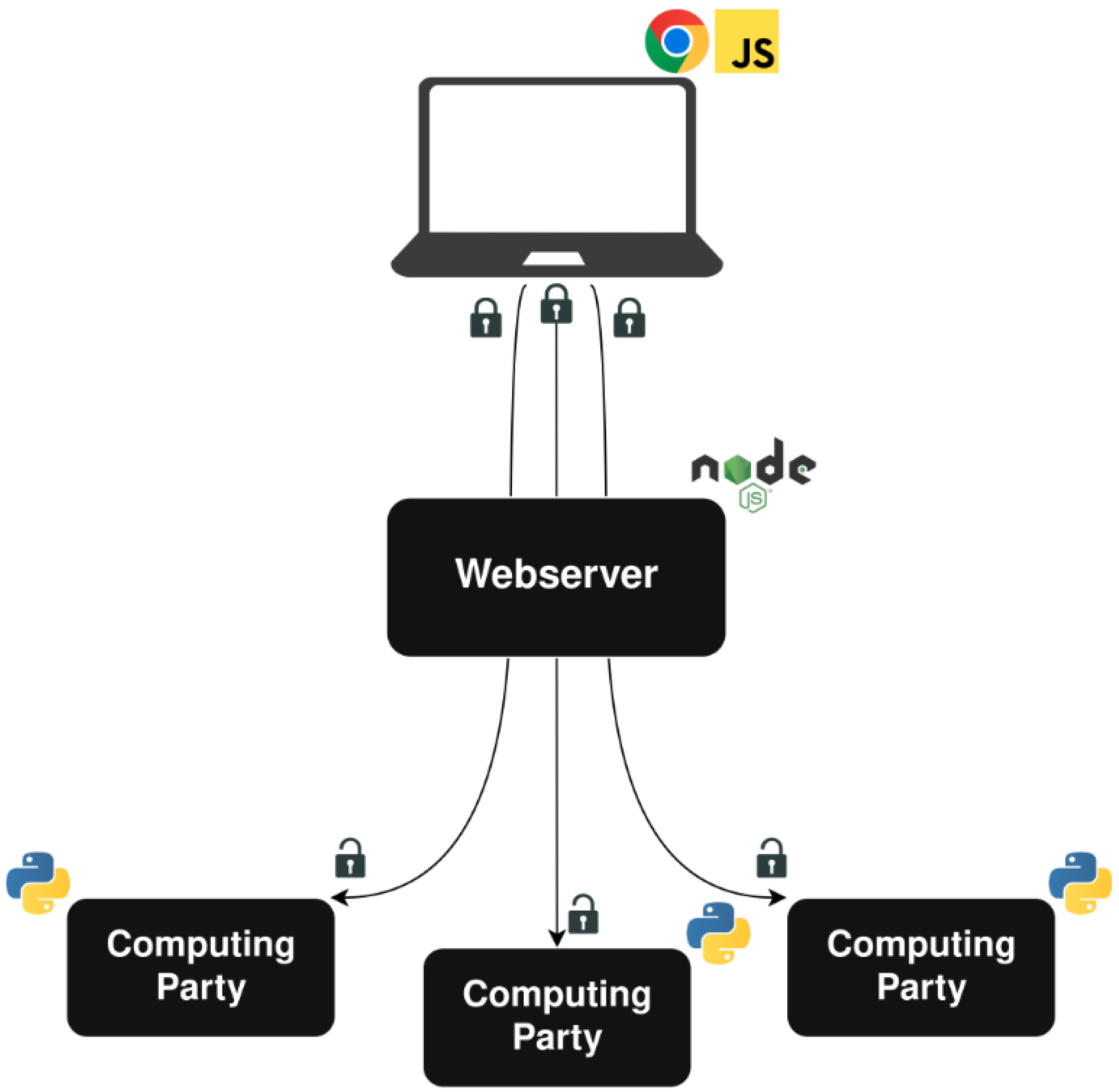
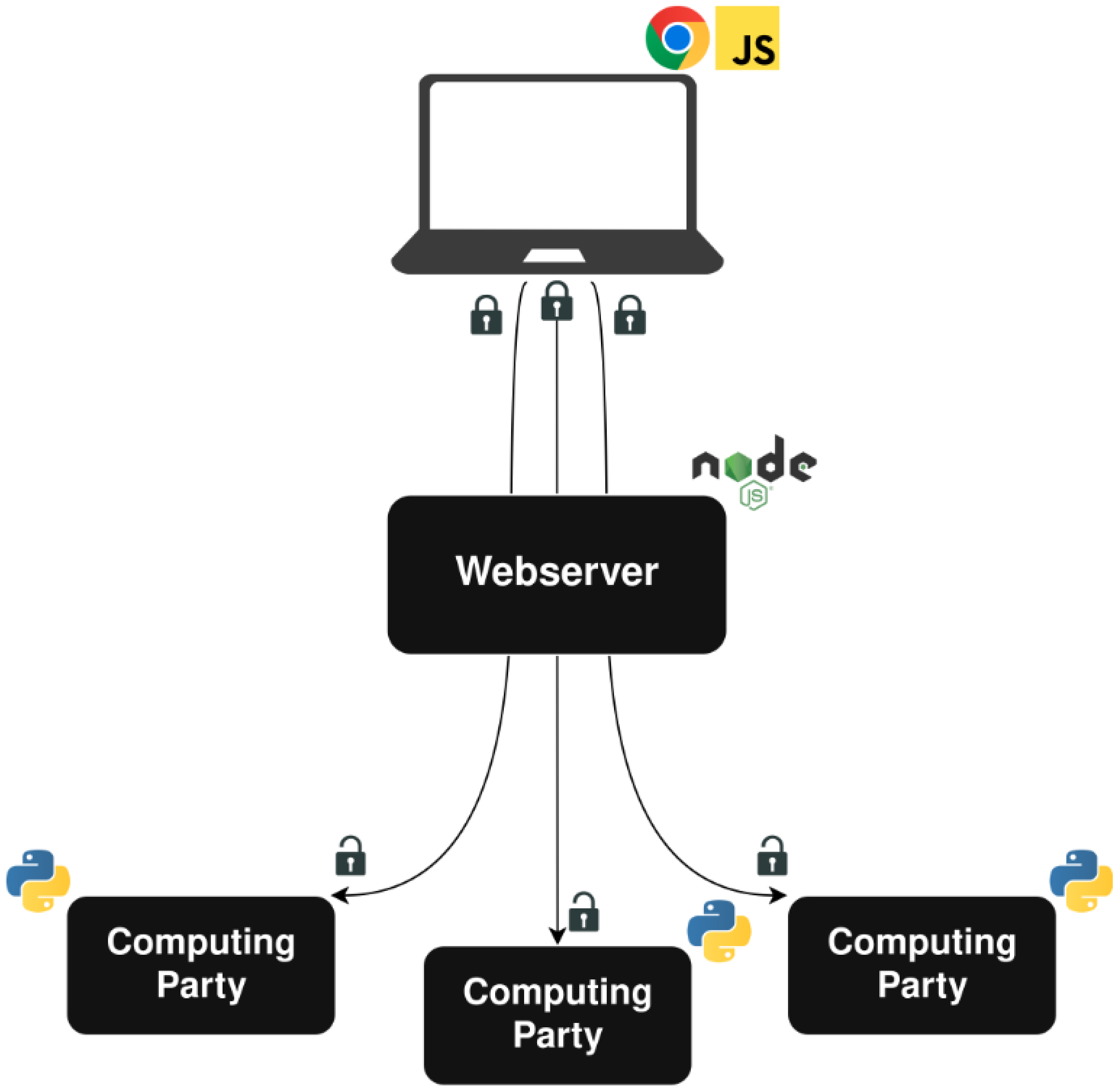


Webserver

- Simplifies interaction with clients
- Collects basic metadata
- Never sees any private data



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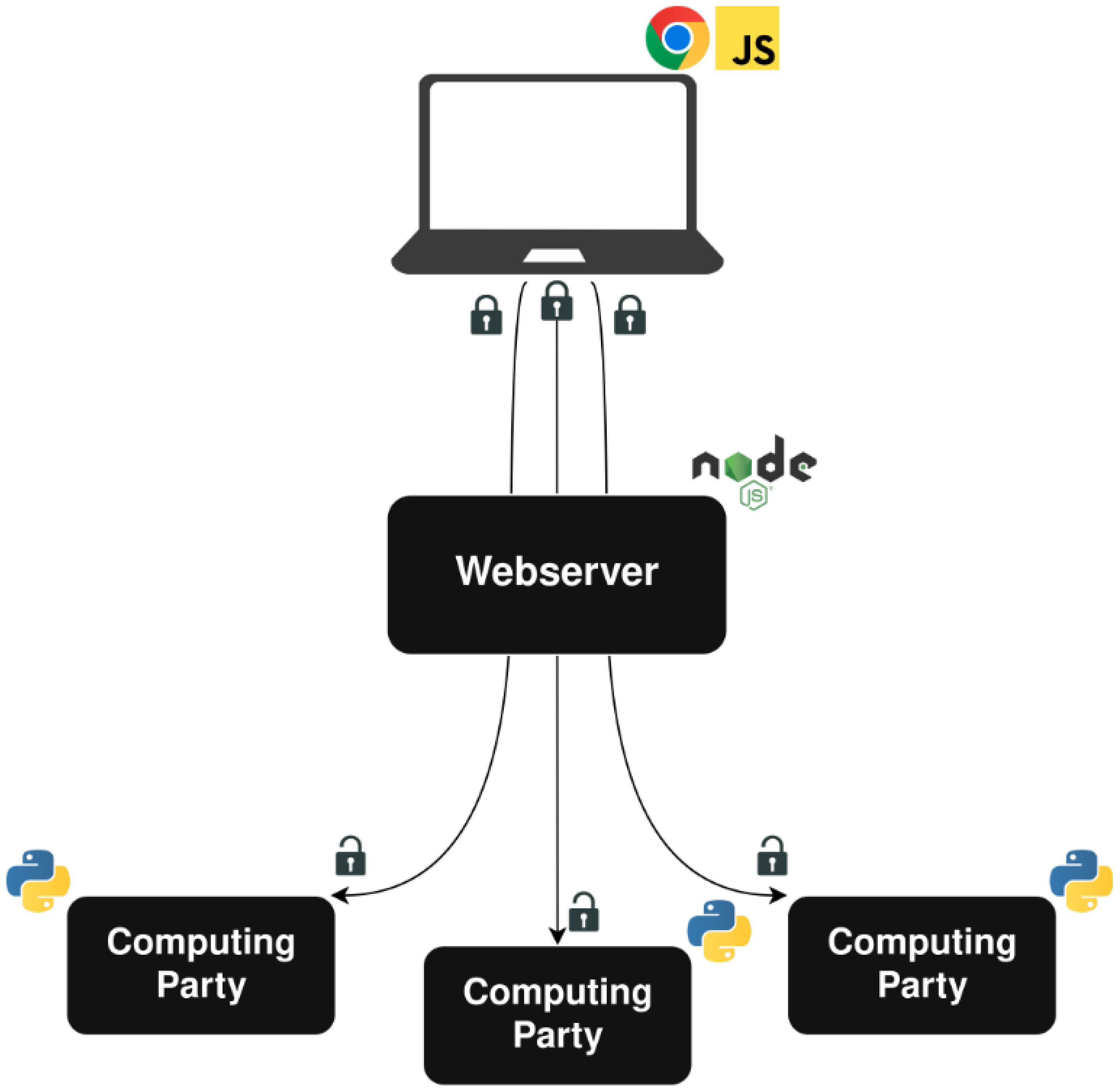

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node
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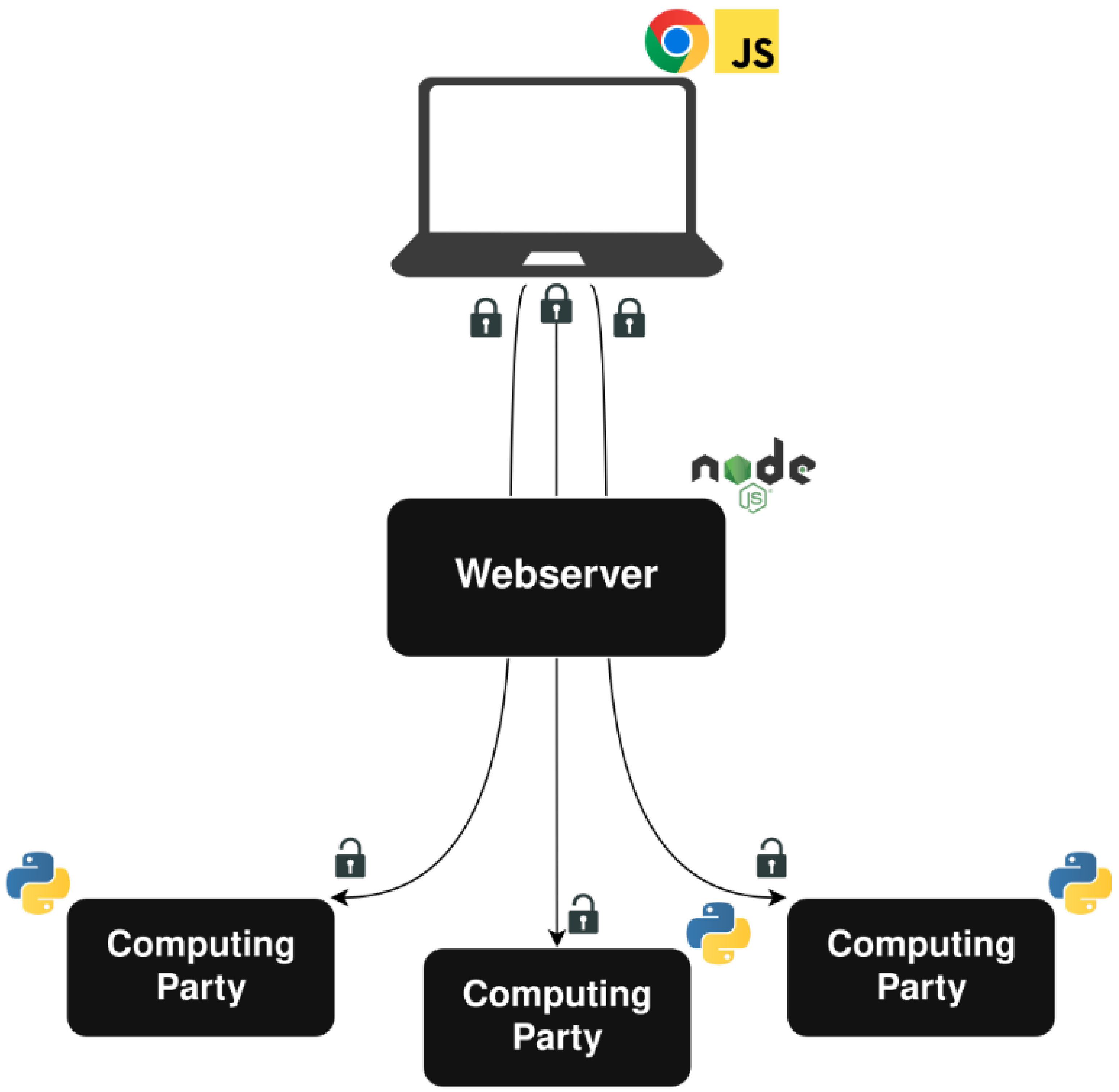
MPC Backend

- Three party computation with an honest majority



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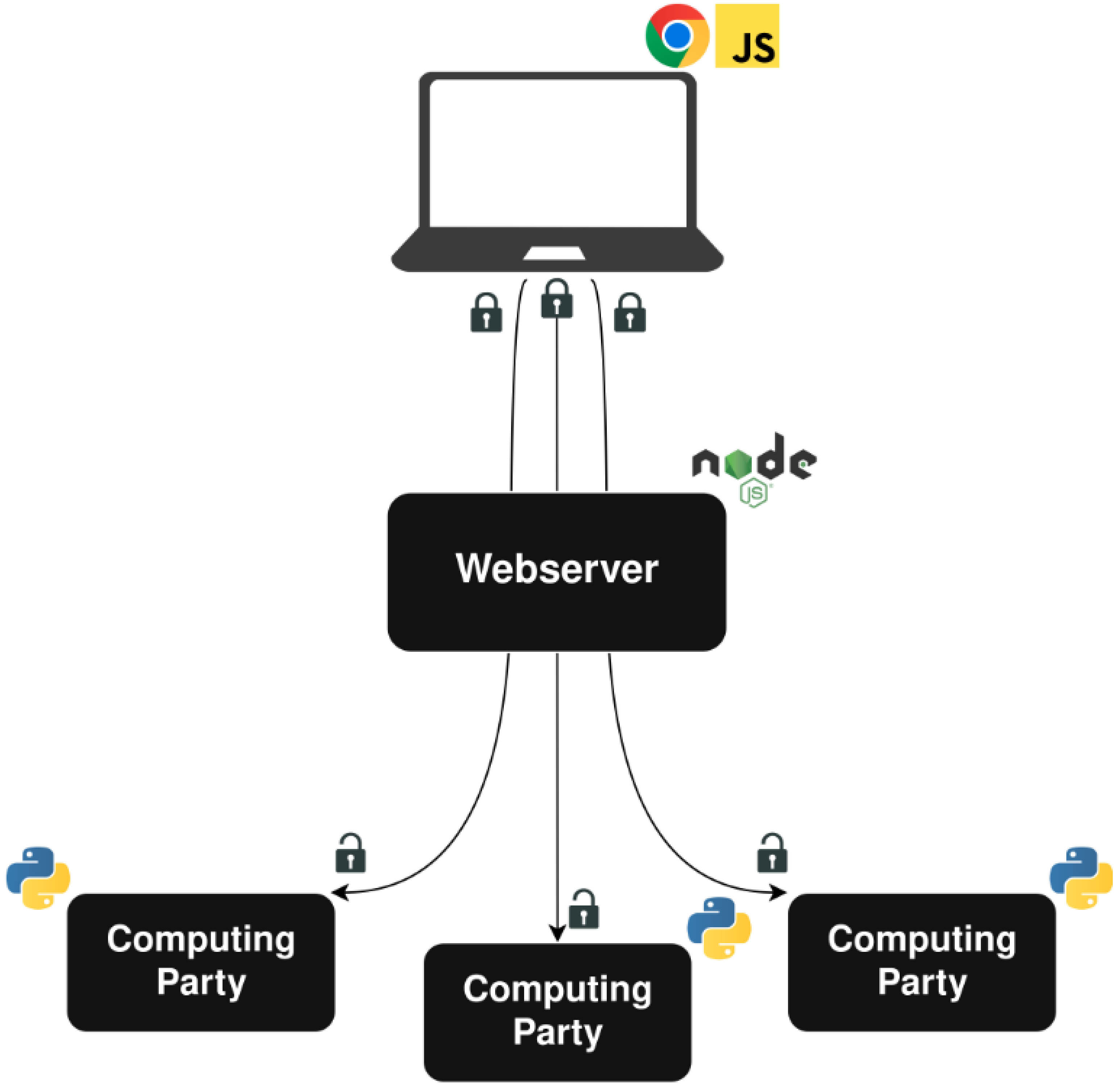
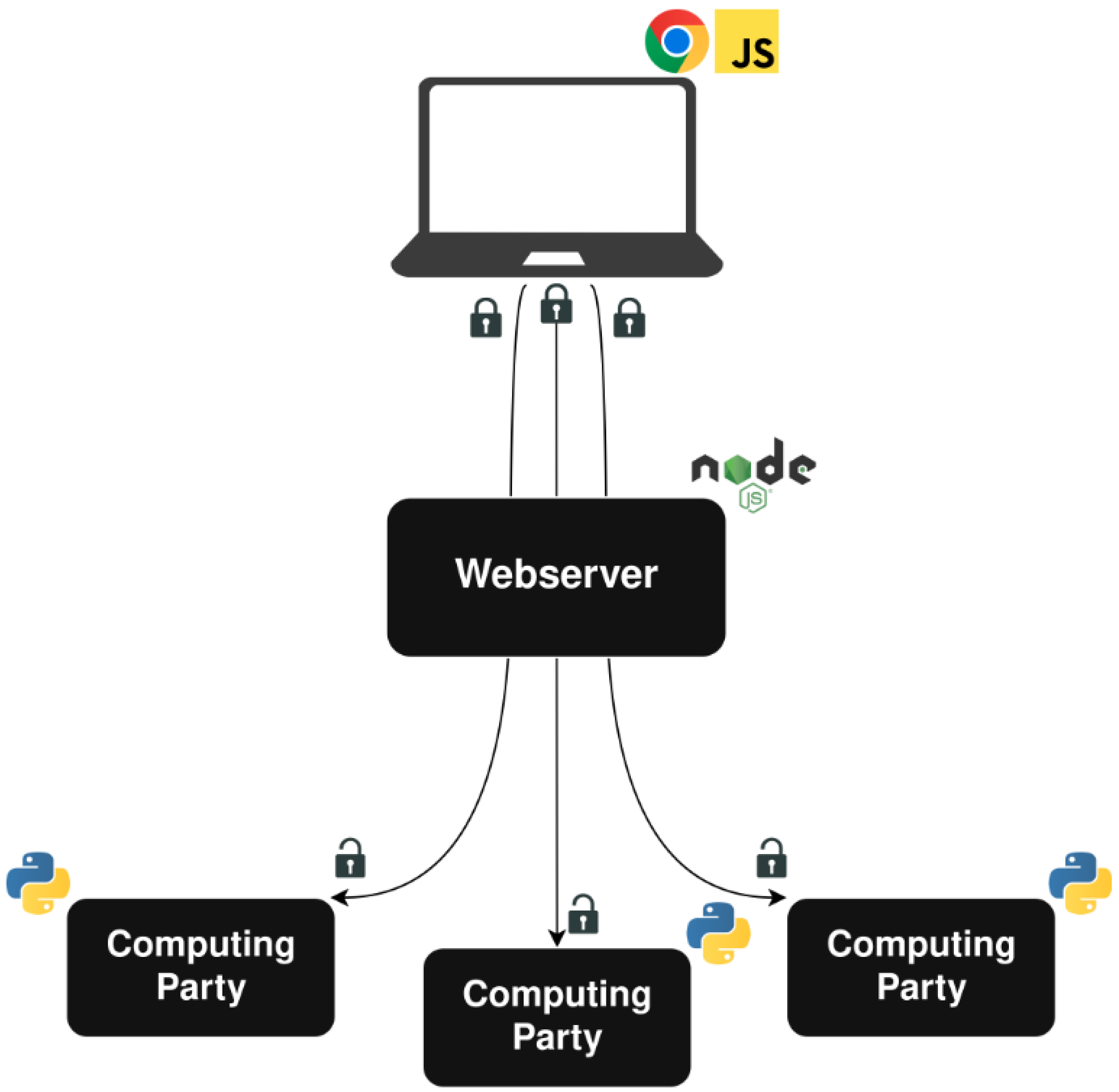


MPC Backend

- Three party computation with an honest majority
- We used and augmented the CrypTen library
- We implemented an algorithm for LLP under MPC



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- The first implementation under MPC of an LLP model
- Mix of out-of-the-box components and custom implementations

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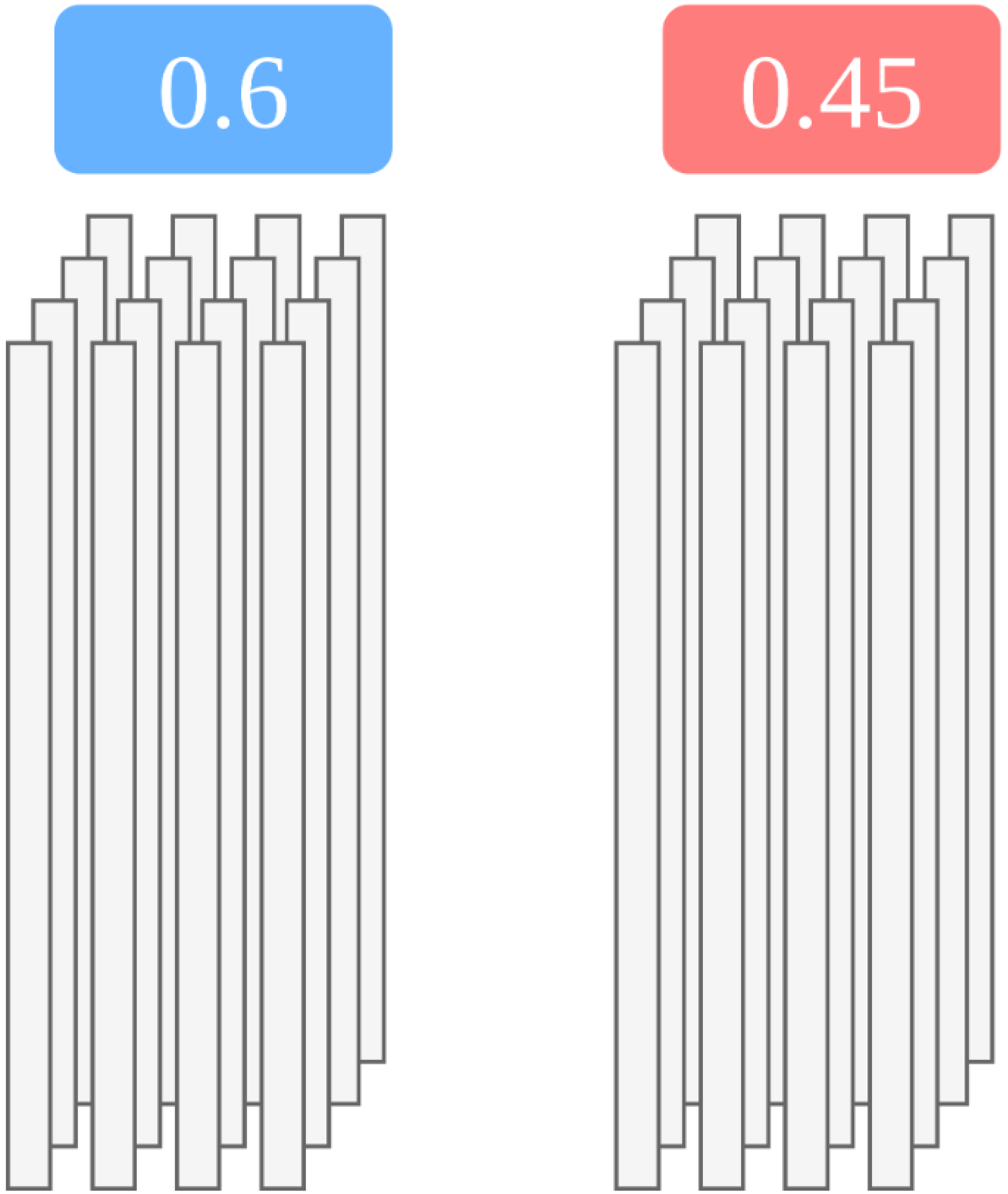

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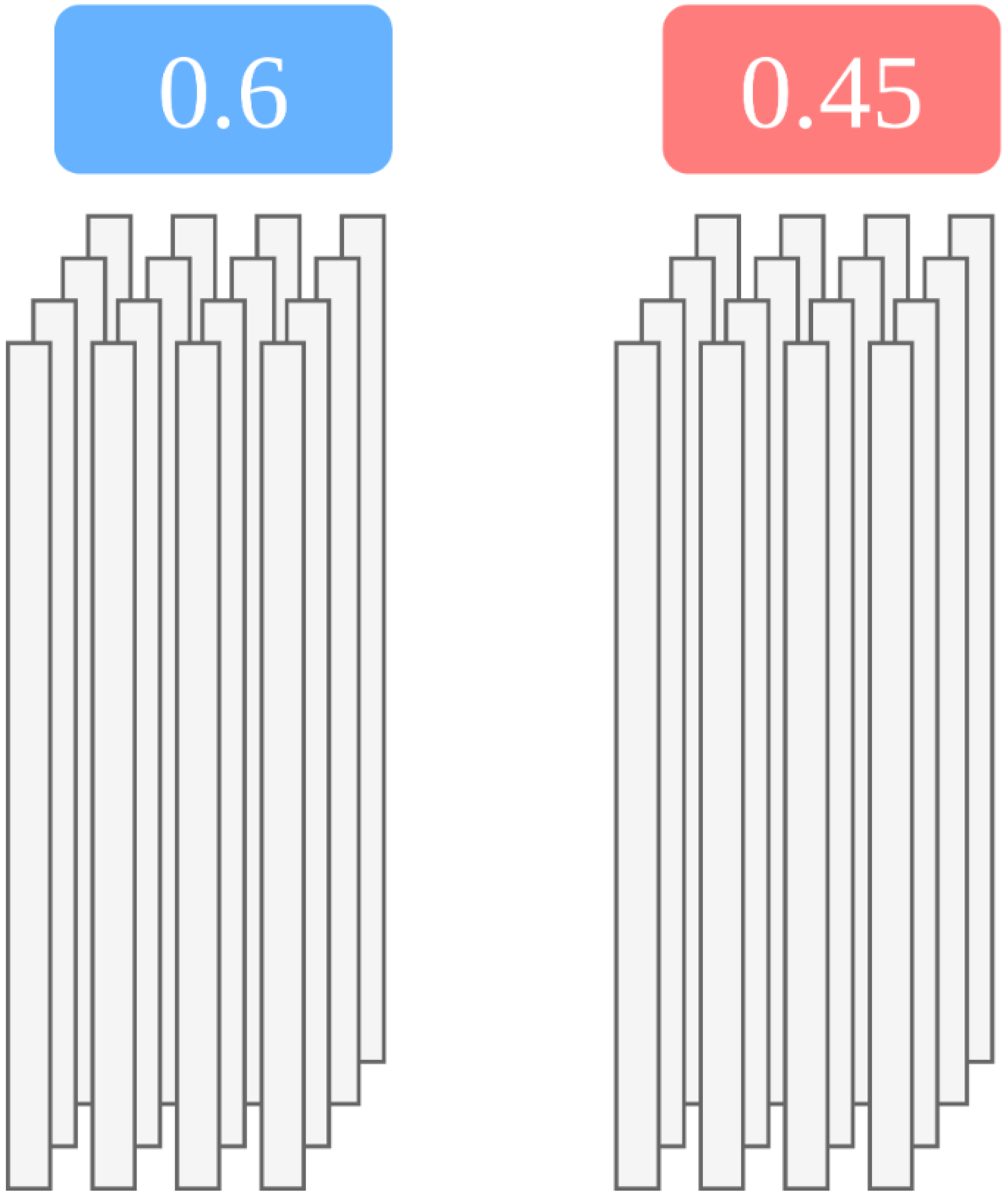
- Train on aggregate ground truth



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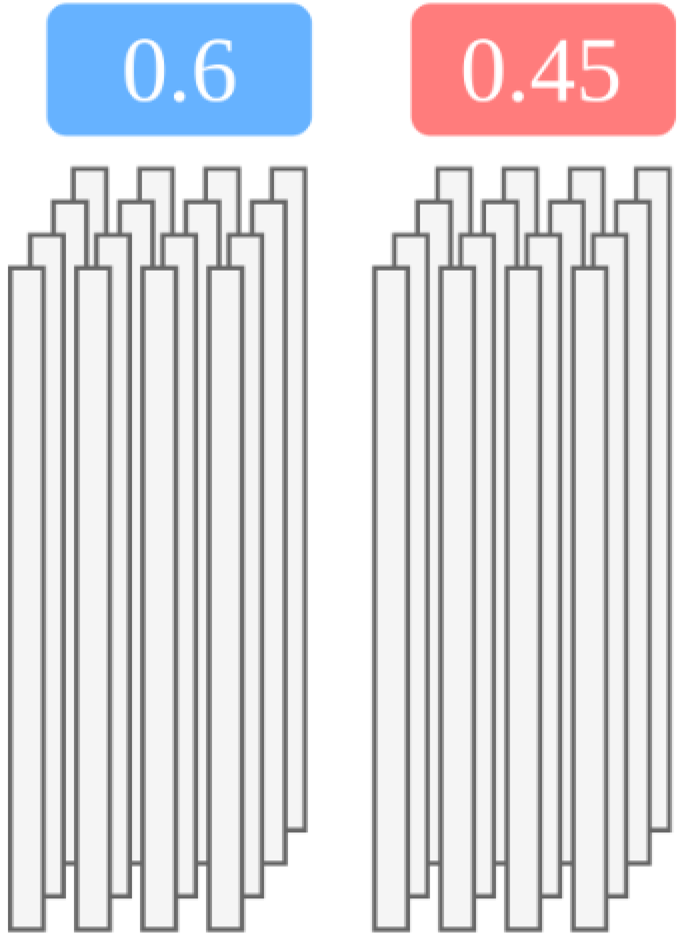

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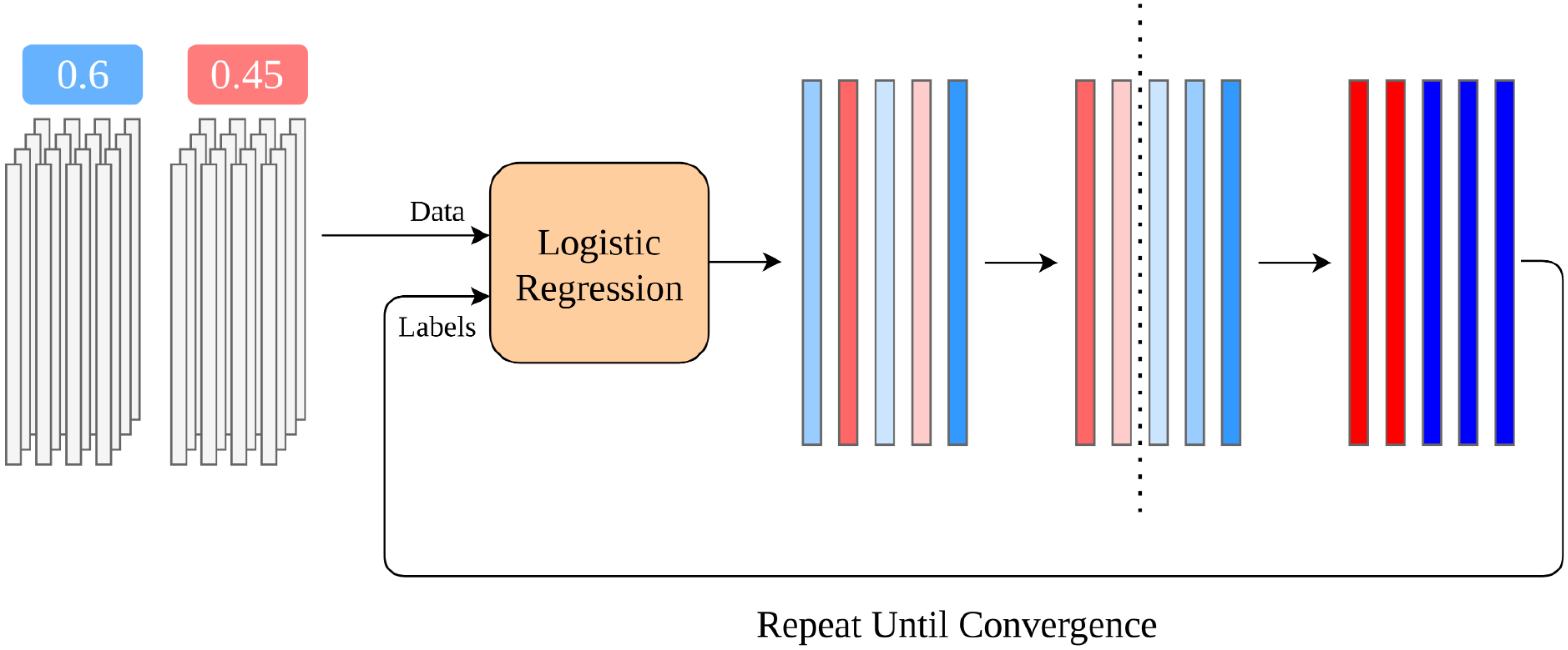

Translating to MPC

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repeat until convergence:
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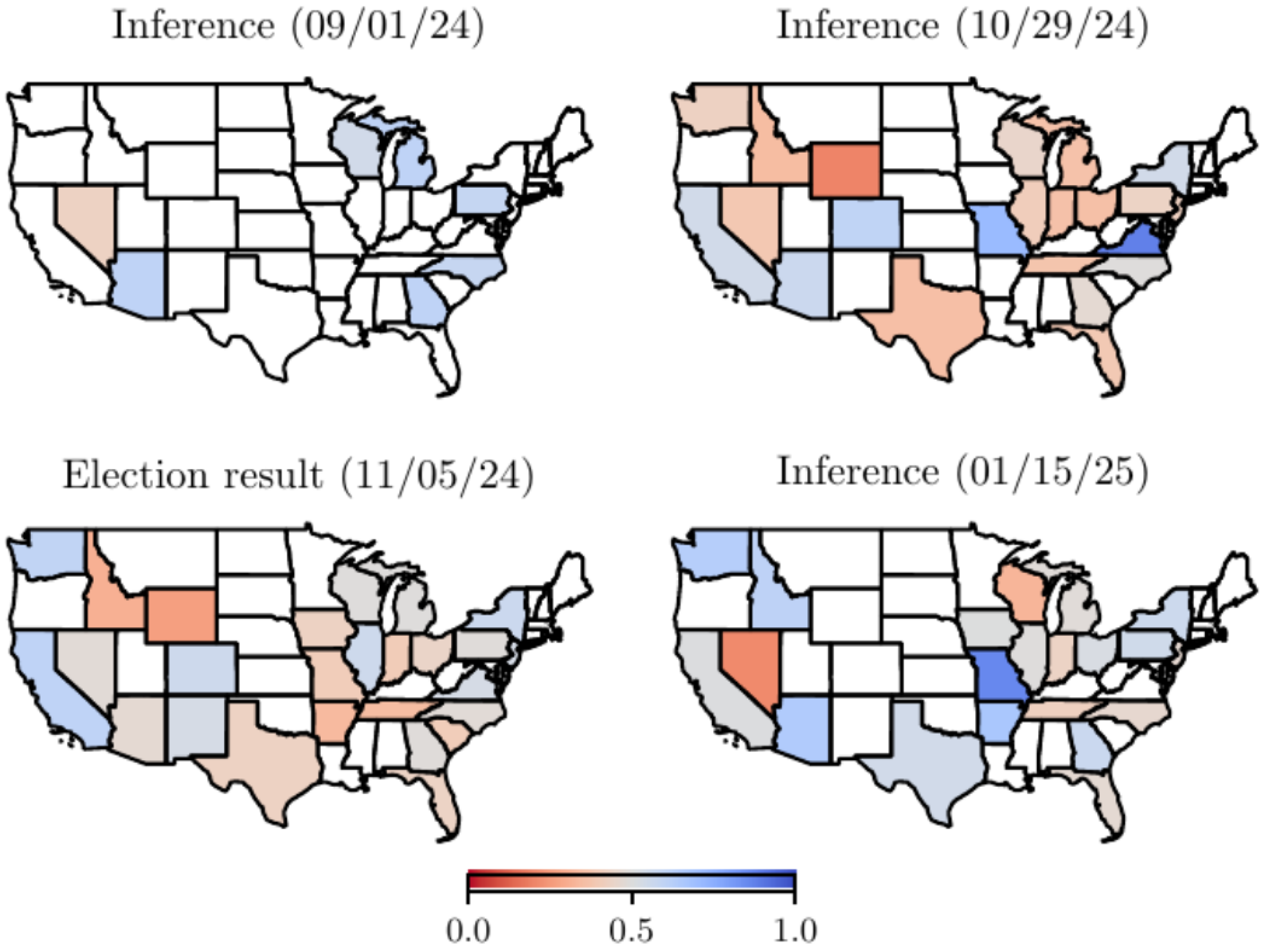
Preliminary Results



Lessons Learned and Future Directions

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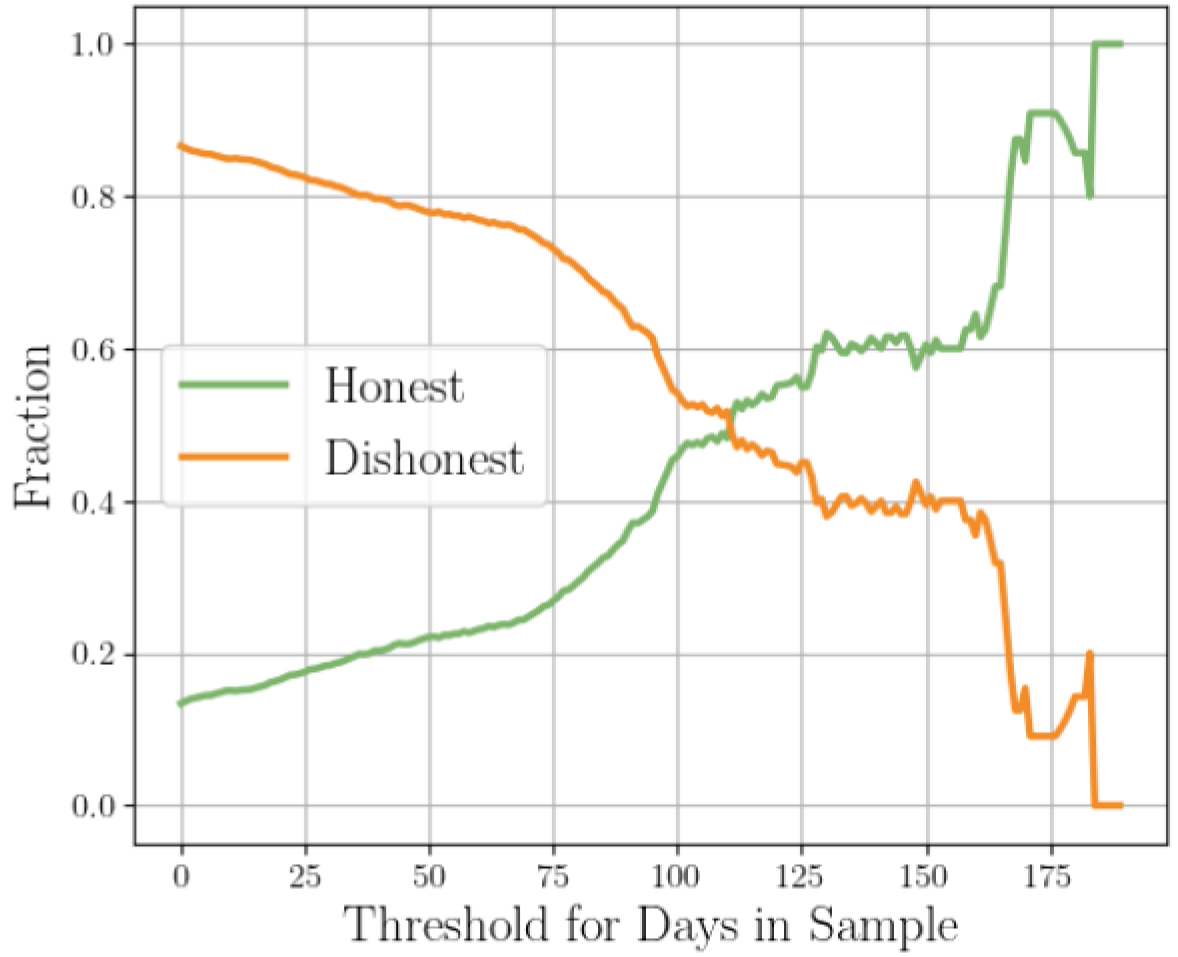

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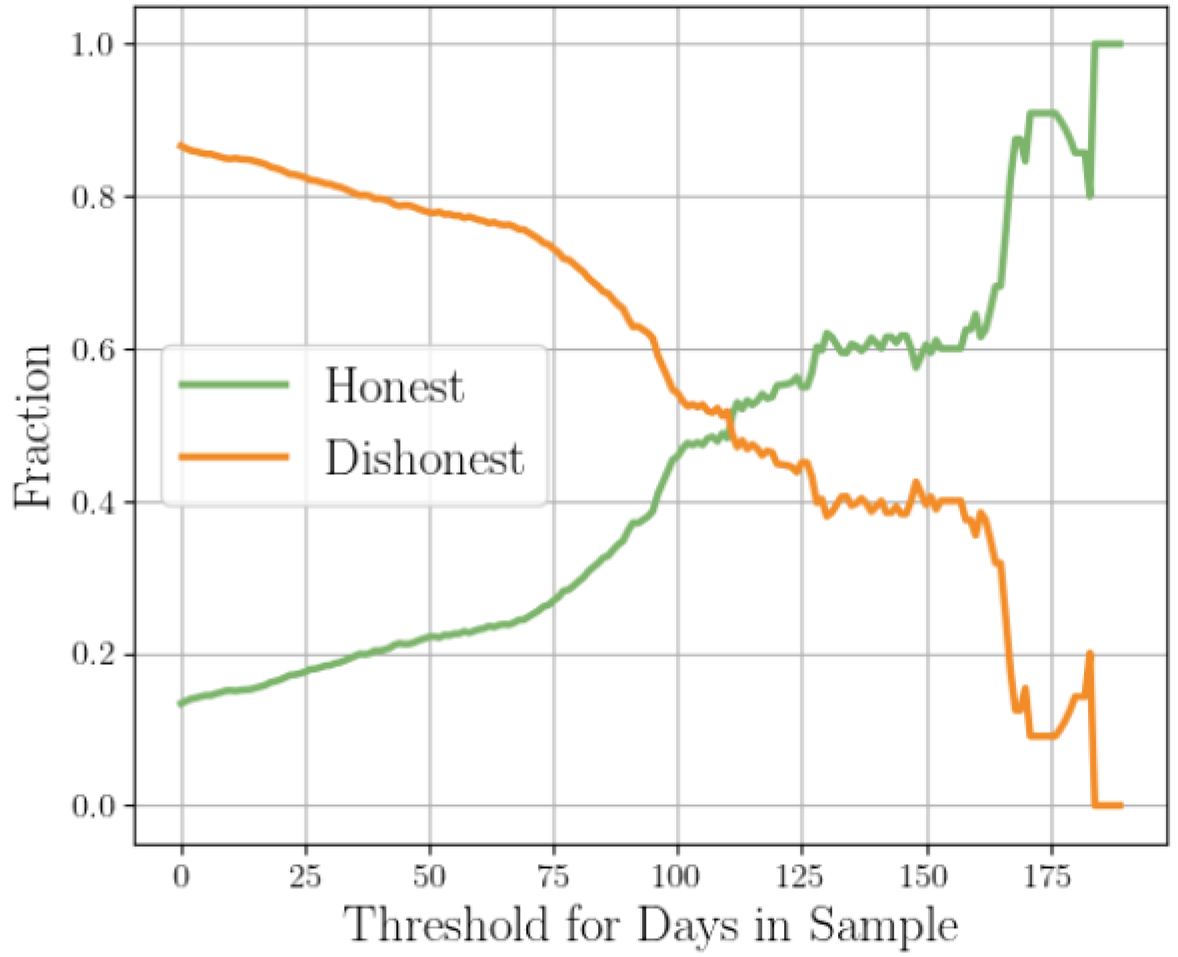

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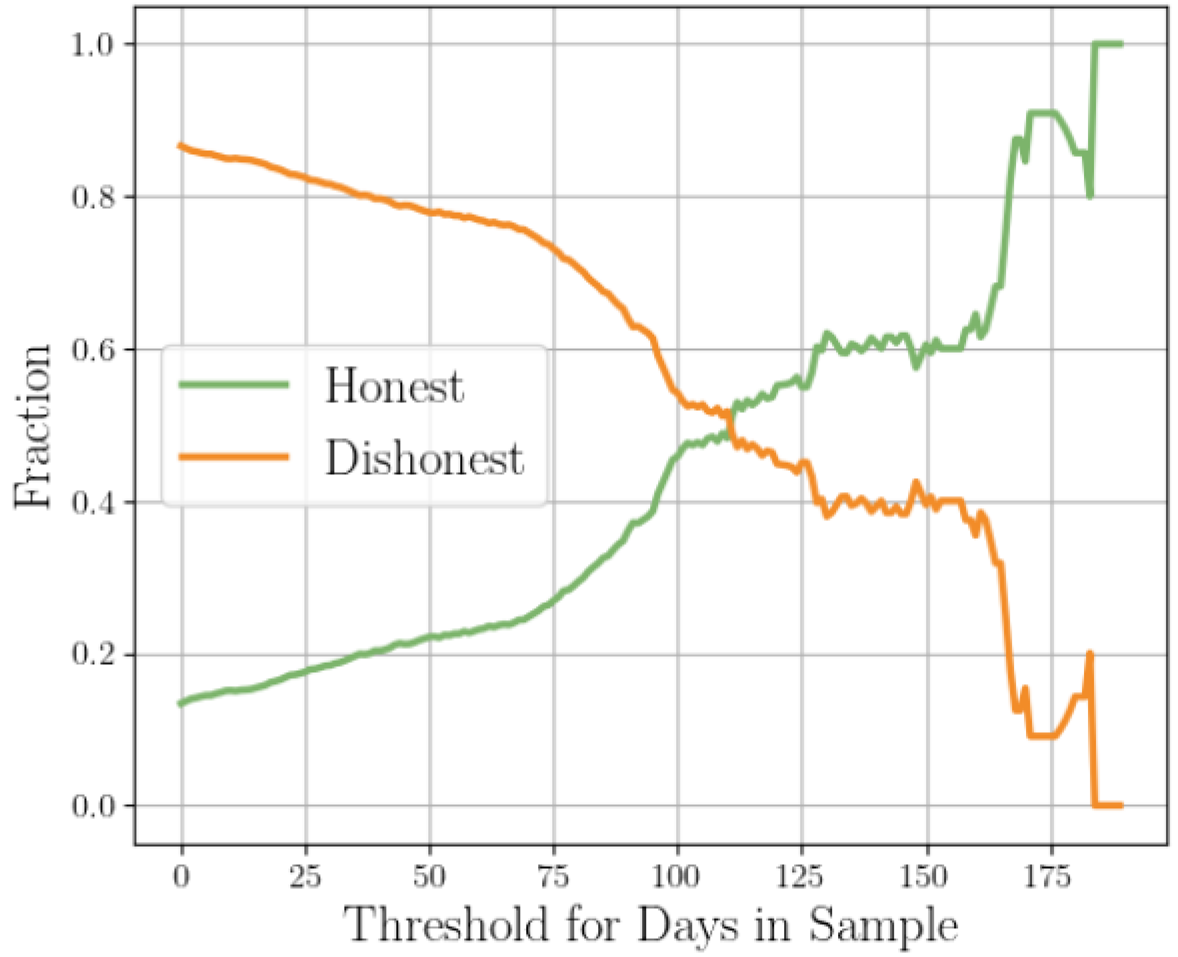

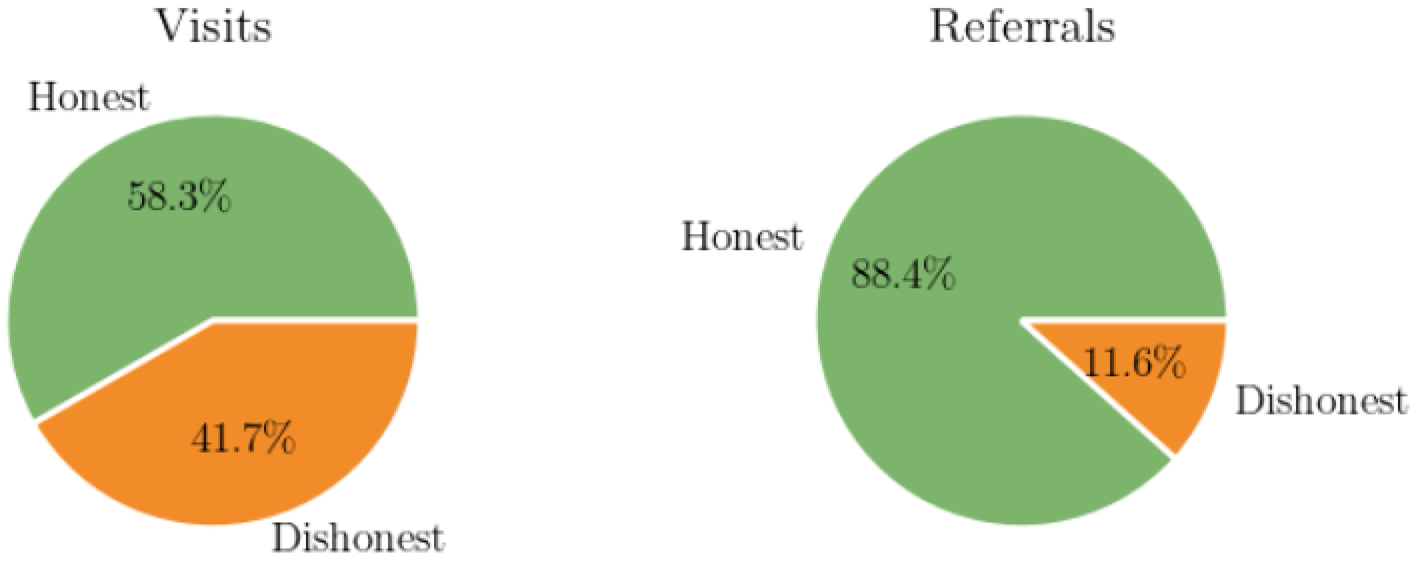

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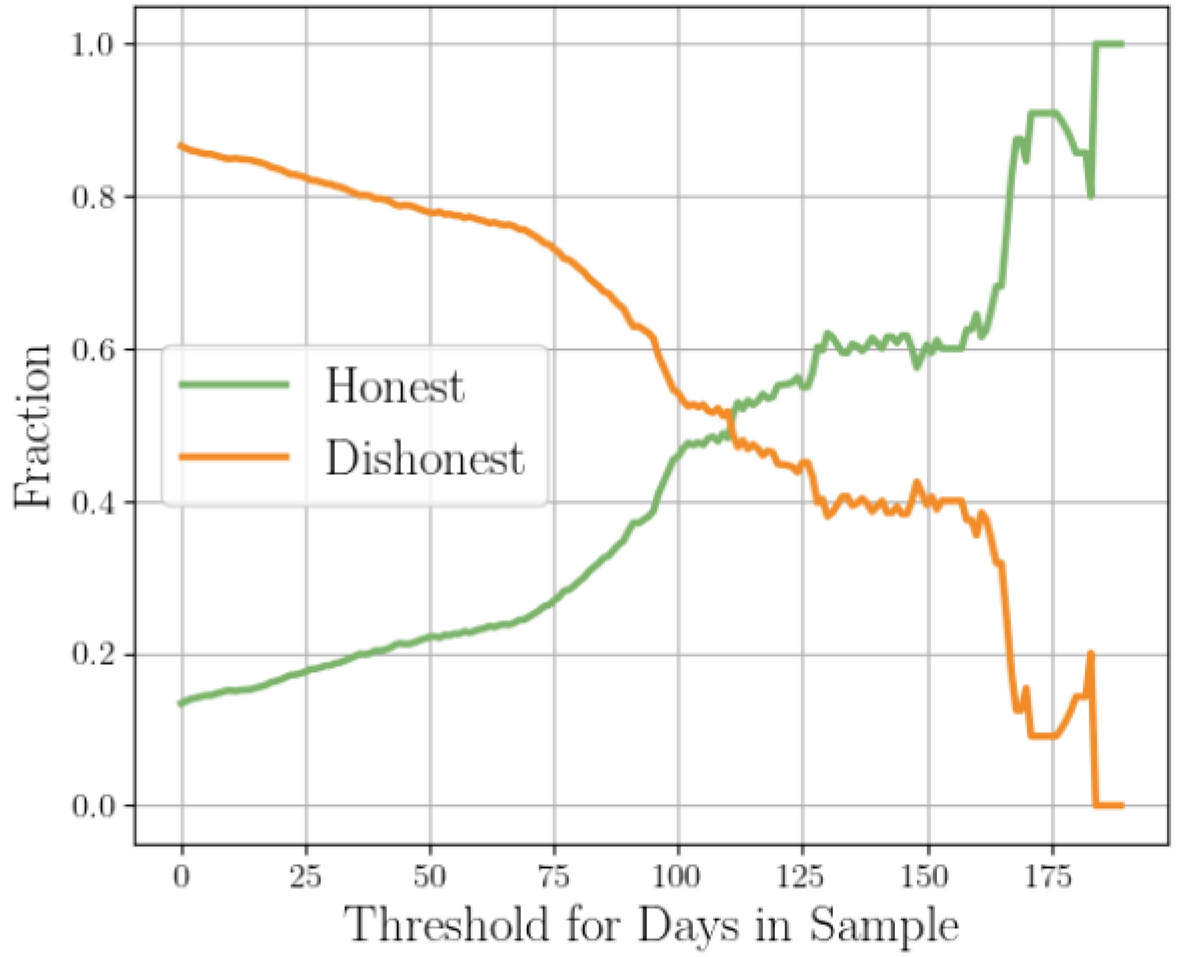

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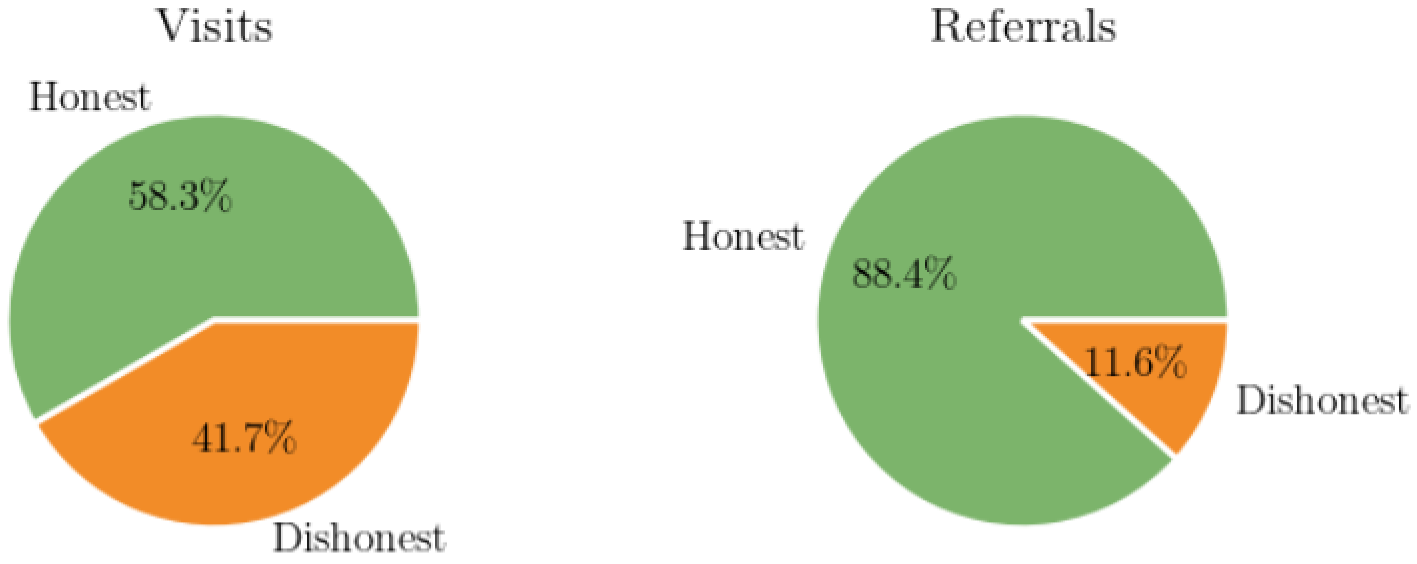

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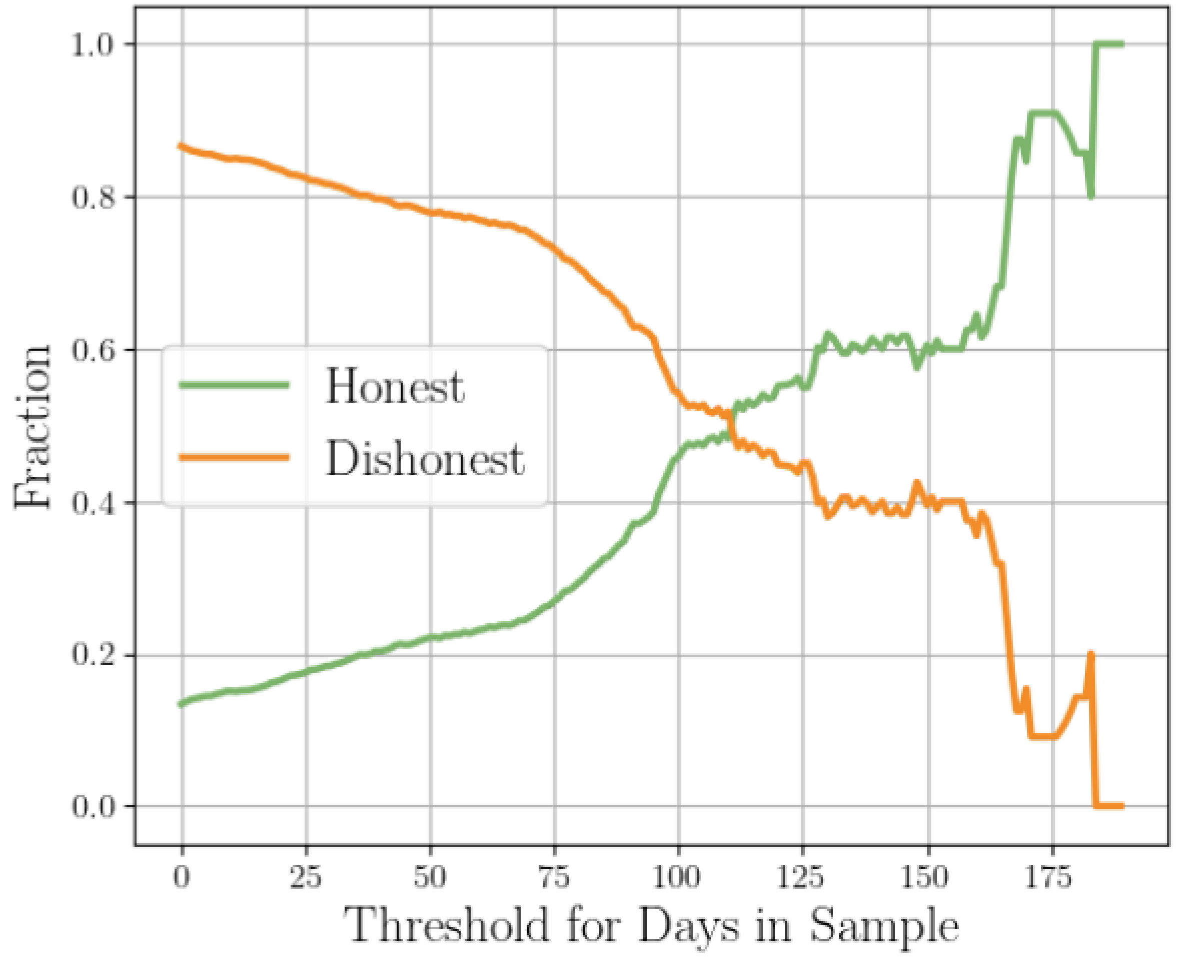

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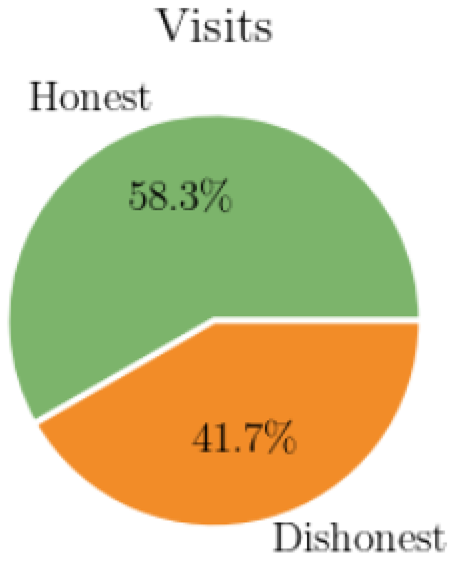

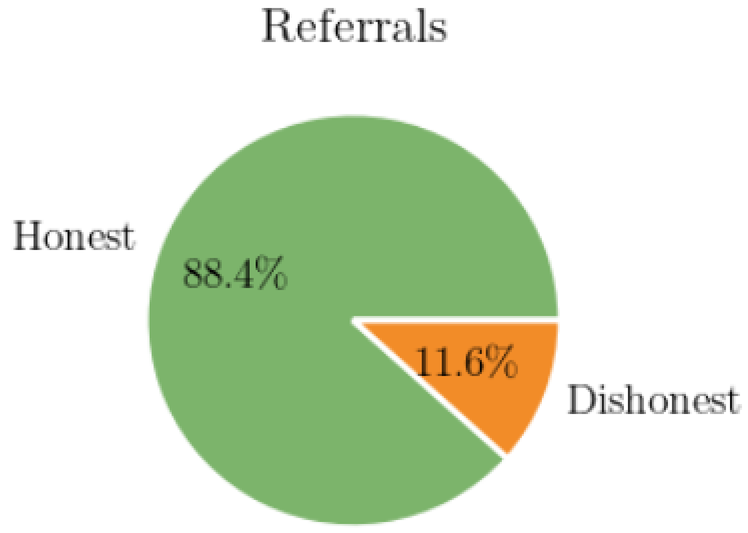

Lesson: Our data is surprisingly robust to dishonest users.

Strengthening the Threat Model

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- AWS as a single point of failure

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- Reduce or eliminate trust in the core computation

Strengthening the Threat Model

- AWS as a single point of failure
- Reduce or eliminate trust in the core computation
- Anonymous payments

Thank You!

sambux@bu.edu

Payment

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- Users are paid and recruited through Mechanical Turk

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amazon mechanical turk

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Communication by email

Data Partition

Data Partition

Data Partition

BOSTON UNIVERSITY

WashU

Input	Web Browsing Data	Registered IDs

Data Partition

	BOSTON UNIVERSITY	WashU
Input	Web Browsing Data	Registered IDs
Output	Aggregation by ID	ID Set per Day

Calculating Payments

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- BU sends all collected tags to WashU
- WashU computes all combinations of users and dates and checks for matches

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- BU aggregates data from same anonymous ID for each week